Export-Import Bank of Malaysia Berhad

STANDARD OPERATING PROCEDURES (SOP) FOR CREDIT RISK MANAGEMENT SYSTEM (CRMS)

Version 2.4 (2023)

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ABOUT THIS DOCUMENT

(A) **Introduction**	This document outlines the objectives and functions of RMD.
(B) **Purpose**	The purpose of this document is to guide staff(s) involved in the credit process, review and loss provision and to serve as a source of reference for all staff.
(C) **Scope of Coverage**	This document shall be circulated to all EXIM Bank staff involved in the credit process, review and loss provision.
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(G) **Other Policies & Guideline**	This document is to be read together with other related internal policies and/or external policies/guidelines issued by the governing authorities which contain specific details. 1. MFRS 9 Financial Instruments Standard 2. MIA FAQs For MFRS 9 Financial Instruments 3. Credit Risk Policy 4. Credit Processes & Procedures 5. ECL Policy 6. 6Sigma CRMS Risk Rating Methodology 2020 7. IFRS 9 in CRMS – Module Description Manual 8. 6Sigma CRMS Risk Rating Methodology 2022 9. EMM Module Manual and CRMS Manual 10. Board Papers of Methodology Revision
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CHAPTER 1: RATING MODELLING METHODOLOGY

1. The CRMS risk rating is built on a Quantitative and Qualitative platform for calculating an Obligor Risk Rating/ Final Risk Rating.
   - The Quantitative data is based on financials, which are events were converted to numbers, and in most part were vetted by third parties such as auditors. Assessment of the quantitative data is reflected in the obligors's Financial Risk Rating.
   - The Qualitative information provides insight and forward-looking indicators on the obligor's Management, Industry and Environment. These are reflected in the obligor's Management Rating, Industry Rating and Environment Rating.

2. The Final Risk Rating is a composite assessment of Quantitative and Qualitative data, grouped into four main categories:
   - Financial Risk Rating (60% of the Final Risk Rating)
     - Historical data are events that have already taken place. **Being a verifiable measurement, the Historical Performance of the obligor carries a high 60% weight of the Financial Risk Rating**, and 25% of the Final Risk Rating. However historical data are just that, past events that may or may not have a bearing on future performance. As such, this sub-category is balanced by other factors such as Future, and more importantly the obligor's Sensitivity to changes in its circumstances
   - Management Rating (15% of the Final Risk Rating)
     - The Management Risk Rating is the first qualitative assessment of the Obligor. The evaluation is made using a series of questions and answers on Management and covering a number of key indicators. The number and nature of the questions are important in gauging qualitative issues and reducing, to a high degree, the judgmental element of the assessment.
   - Industry Rating (15% of the Final Risk Rating):
     - Studies conducted by Citibank, JPM Chase and others have shown that economic recessions tend to increase the number of defaults. The rate of increase depends on many factors, including management expertise and the obligor's tolerance to changes in its circumstances.
     - As such, although Industry and the Environment play a role, their overall impact on the obligor's probability of default is limited. In the case of the CRMS, these are set at 20% of the total rating, divided equally.
   - Environment Rating (10% of the Final Risk Rating)
     - The Environment Risk Rating is the third qualitative assessment of the Obligor. The evaluation is made using a series of questions and answers on the Obligor's environment. As with the other qualitative questions, the number and nature of the questions are important in gauging qualitative issues and reducing to a high degree the judgmental element of the assessment.

3. Financial Risk Rating
    - Within this category, the distribution of weights is fifty percent (50%) for the assessment of Historical Financial Data, while the remaining balance split between Future (33.33%) and Sensitivity (16.67%).
    - The success of an obligor's business model revolves around the **ability to generate Net Operating Cash Flows (NOCF)**, given that it is used to settle interest charges and short-term obligations towards banks.
    - The two main drivers of NOCF are EBITDA and Cash Cycle, which together help in generating (or consuming) NOCF for the company.
        - Earnings before Interest, Taxes, Depreciation & Amortization (EBITDA) represents the company's operating profitability. All other things equal, the higher the EBITDA margin, the higher is the profitability cushion that allows for greater absorption of changes in working capital accounts (i.e. changes in the Cash Cycle), and therefore the lower is the risk.
        - The Cash Cycle represents the period it takes a company to collect cash from its operations. The greatest impact on NOCF from the Cash Cycle is either the company is witnessing large swings in its Cash Cycle or the Cash Cycle is greater in percentage terms than the EBITDA margin thereby consuming more cash flow than is generated.
    - **NOCF volatility with 15% weightage can be justified as:**
        - NOCF volatility is an important indicator of the reliability of the operating cash flow streams of the borrower which in most cases are relied upon for loan repayments (incl. interest servicing).
        - Low volatility in NOCF is preferred compared to high volatility given that low volatility indicates a steady cash flow streams generated by the operating businesses of the borrower. This is good from the credit profile perspective.
        - Hence, a 15% weightage is justified as the CRMS rating model is cash flow centric
    - The next level of analysis is the **measurement of sufficiency**. Once they are generated, it is important that these cash flows are sufficient to cover the obligor's short-term obligations towards the banks. The three main parameters and their weights are Debt Service Coverage Ratio (DSCR) Interest, DSCR Short-term Bank Debt, and DSCR Total Bank Debt. **6 Sigma research suggested that DSCRs play a crucial role in predicting default and justified with higher weightage.** As such, if any of the DSCR levels were very low, then the Financial Risk Rating increases to at least 6.5.
        - DSCR Interest: Measures the ability to cover bank Interest Obligations from NOCF. This is an important criterion as not being able to cover interest indicates a high probability of default (despite having the ability to cover it from other sources such as additional bank debt).
        - DSCR Short-term Bank Debt: Measures the ability to cover short-term bank repayments. Obviously, the obligor has to have the ability to cover its short-term commitments. As these are short term in nature, then NOCF should be relatively sufficient to cover them over the forthcoming fiscal year. Mismanagement of bank facilities in this respect would provide for a higher rating.

    - DSCR Total Bank Debt: Measures the ability to cover total bank debt balances. This measurement reflects the obligor's ability to meet total bank debt within a relatively acceptable average period of 3 years.
- While reliance on debt financing is a useful method for shareholders to increase the return on capital, it also places more pressure on meeting commitments. The two main parameters that measure a company's reliance on debt financing are Leverage Total Liabilities: Measure of total Liabilities vis-a-vis equity (1%), and Leverage Bank Debt: Measure total bank debt vis-a-vis equity (1%)
- **Projections** represent future events that are expected to take place. Although projections are invariably inaccurate in absolute terms, they are expected to provide a picture of the obligor's future cash generating ability. To counter-act any bias introduced by users to enhance future performance, projections are balanced by Sensitivity Analysis, which measures volatility and sensitivity (addressed later All projected criteria will carry the same weight as historical).
- **The Sensitivity** represents the measurement of vulnerability to changes in the obligor's future circumstances (major business drivers), and impact on its future cash generating ability. The higher the tolerance to change, the lower is the rating and vice versa. The major business drivers are as follows, with equal distribution of weights into Sales, Cost of Sales I Sales, SG&A/ Sales, Receivable Aging, Inventory Aging, and Payable Aging.
- **The weightage allocation in all the financial ratios are tested on regular basis by 6 Sigma and justified by the high accuracy of Gini Coefficient.** The result of the Gini Coefficient is documented in Chapter 7.
- Refer to Appendix A and B for details.

4. Management Rating

- It covers the following six indicators with equal distribution of weights into Character, Quality I Skills, Organization Structure, Depth and Succession, Risk Management, and Relationships with Other Banks
    - Character carries a 5% weight of the Final Risk Rating and is an assessment of the following:
        - Management's integrity
        - Management's reliability
        - Management's level of commitment towards the company
    - Quality and Skills (5% of Final Risk Rating). It is an assessment of Management's knowledge and expertise in the following key areas:
        - Operational Expertise: Degree of operational know-how.
        - Financial Control and Expertise: Degree of financial management & control of the operation.
        - Marketing Skills: Knowledge of products & services, customer needs, and competition.
        - Professionalism: Quality of interaction with third parties.
    - Organization Structure (5% of Total Risk Rating) covers the following key areas:
        - Hierarchy: Structure and degree of delegation in the decision-making process within the organization
        - Strategic Management: Extent to which the company's vision, mission and objectives are communicated across the organization
        - Human Resources: Management of employees within the organization
    - Depth and Succession (5% of Total Risk Rating) defines the degree of depth within the company and the assignment of successors across the management hierarchy.
    - Risk Management (5% of Total Risk Rating) defines management's ability to identify risks associated with the company and manage or mitigate its exposure to such risks. These risks are categorized into four key types:
        - Market Risk
        - Credit Risk
        - Operational Risk
        - Local I Government Regulations
    - Relationships with Other Banks (5% of Total Risk Rating) define Management's ability to diversify its bank facilities and acquire the necessary financial support for its business.
    - **The qualitative questions are designed to be answered in a forward-looking manner by the Bank. These questions are made granular (usually carry the same weights), so they spread across a large base. They are also objective and hence subject less to volatility in answers.**
    - The weightage has been cut down from the original 30% to the current 15% because the Bank still observes the forward-looking elements in the answers with each credit proposal review.
    - Refer to Appendix B & C for details.

5. Industry Rating

- The Industry Risk Rating is the second qualitative assessment of the Obligor. The evaluation is made using a series of questions and answers on the Obligor's Industry. As with Management, the number and nature of the questions are important in gauging qualitative issues and reducing to a high degree the judgmental element of the assessment.
- Industry carries a weight of 10% of the total Risk Rating and covers the following three indicators: Industry-Specific Assessment, Company-Specific Assessment, and Sources of Information.
- Industry-Specific questions and their results are applied to all Obligors within the same industry. They tackle the following key areas:
    - Growth Trends and Development Phase: Stage of development of the Obligor's industry and current / expected growth trends.
    - Competitive Environment and Entry Barriers: Competitiveness of the industry and how easy / difficult it is for new companies to enter and compete within this industry.
    - Capital and Technology Requirements: Degree to which investments in capital and technology are required.
    - Regulations: Regulations imposed on the industry by local governments and the extent to which such regulations change.
- Company-Specific questions cover performance of the Obligor's company within the industry. They cover the following key indicators:
    - Production and Facilities: Condition of the company's production facilities.
    - Market Share and Competitiveness: Company's share of the market and how competitive the operation is relative to other companies within the same industry.
    - Market Image and Reputation: Company's image with its customers, suppliers and employees.
    - Concentration Risks: Degree of concentration in the customer base, supplier base, and product base.
    - Technology: Extent to which technology is invested in and updated as required.
    - Other SWOT: Extent of reliance on resources and raw materials subject to high volatility
- Sources of Information section validates the information used to answer the previous questions. It assesses to what extent the Relationship Manager executed the required research to arrive at the above conclusions.
- Refer to Appendix B & C for details.

6. Environment Rating

- The Environment carries a weight of 10% of the total Risk Rating, and assesses the environment in which the Obligor operates at the following three levels (with equal distribution of weights):
    - Economic: Macro-economic environment of the Obligor's country of operation
    - Political: Political stability of the Obligor's country of operation
    - Contagion: Degree to which the Obligor's environment is impacted by the geopolitical environment of neighboring countries
- Refer to Appendix B & C for details.

CHAPTER 2: INTERNAL RATING GENERATION

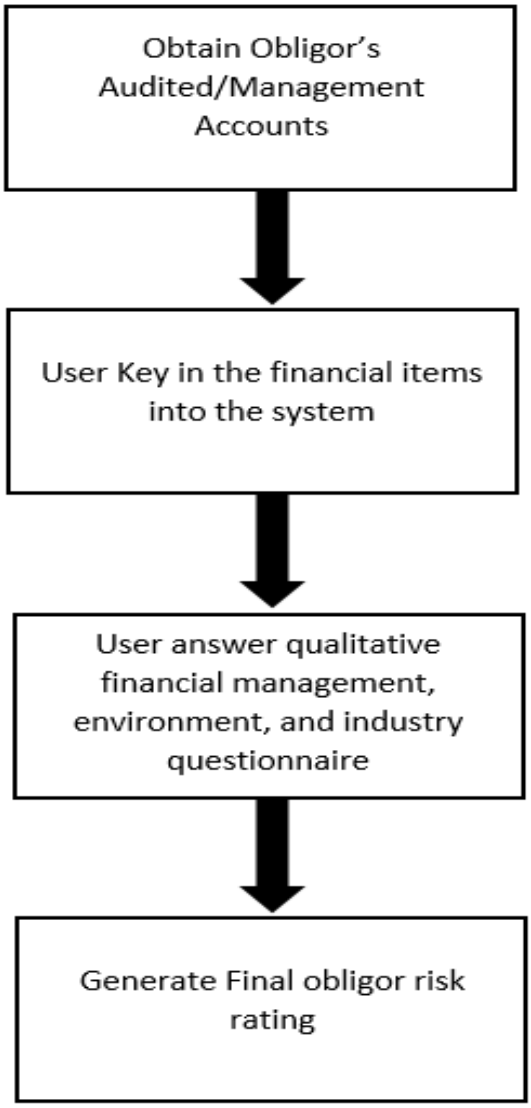
1. CRMS Rating Generation

- A Risk Rating is derived for each criterion by multiplying each parameter score with its weight. The scores are based on a scale suitable for the type of parameter in question. As an example, the Sales Growth scale ranges from low of +10% to a high of -10%.
- The Scale Range applied to each parameter is judgmental and is initially provided by CRMS. However, it is based on studies and research conducted by CRMS since 1983 on what constituted acceptable ranges for each item in the ultimate calculation of risk. These ranges have since been verified through the application of the CRMS on regional cases and have withstood close scrutiny by 6 Sigma and system users alike (see 6Sigma CRMS Rating Methodology February 2021 / Validation section). Refer to Appendix B.
- In any case, users will ultimately have the ability to manage these scores if users have sufficient evidence to warrant changes. Changes are more likely to happen with weights instead, as these are more readily verifiable through regression analysis and similar analytical tools.

2. Target Market Risk Acceptance Criteria (TMRAC)

- The TMRAC module allows the Administrator to set rules to govern the quality of credits being processed through the system. The system automatically checks obligor data against these rules and allocates a number of deviations to it.
- Hence, TMRAC represents the minimum conditions under which the Bank is prepared to enter transactions bearing an element of credit risk.
- The TMRAC is embedded in the Credit Risk Management System (CRMS) and must be applied to each credit proposal. The specific TMRACs for different 3 target markets are as follows:
- General TMRAC – applicable to all EXIM's customers except for the SME customers and customers who fall under the specific product programme.
- TMRAC for SME – applicable to SME customers.
- TMRAC for product programmes – applicable to the specific product programme.

3. Credit Process Flow



CHAPTER 3: EXPECTED CREDIT LOSS

1. MFRS9
   - MFRS9 Financial Instruments was issued by the Malaysian Accounting Standards Board and effective since 2018. It provides the accounting guidance for classification and measurement of impairment.
   - An expected credit loss (ECL) is the expected impairment of a loan, lease or other financial asset based on changes in its expected credit loss over a 12-month period or its lifetime.
   - A forward-looking approach requires the Bank to update and recognize ECL for financial asset or loan from the initial acquisition or origination date.
   - Table below is the extract from MFRS9 that related to forward-looking element in impairment consideration.

2. Forward-looking
   - All the statistical techniques that are used are based on numbers that have already taken place, including incidents of default. The CRMS overcome this issue by including the forward-looking elements in the rating methodology as follows:
     - The cash centric nature of the model calculates the client's ability to generate cash flows, and assesses whether this ability is strong, weak or non-existent.
       - EBITDA's relationship with the Cash Cycle, a cash centric methodology, and
       - DSCRs based on NOCF versus interest, short and total bank debt.
     - The results of the analysis is embedded in the PD rate used in the rating. A client that has a weak business model is unlikely to change in future unless drastic change to it is made.
   - As such, in almost all cases, this model is forward-looking by itself.

Section	Description
B5.5.1	• Typically, credit risk increases significantly before a financial instrument becomes past due or other lagging borrower-specific factors (e.g. a modification or restructuring) are observed. • Hence, when **reasonable and supportable information that is more forward-looking** than past due information is available, it must be used to assess changes in credit risk.
B5.5.17 (f)	• The following non-exhaustive list of information may be relevant in assessing changes in credit risk: Existing or forecast adverse changes in business, financial or economic conditions that are expected to cause a significant change in the borrower's ability to meet its debt obligations, such as an actual or expected increase in interest rates or an actual or expected significant increase in unemployment rates.
B5.5.17 (g)	• The following non-exhaustive list of information may be relevant in assessing changes in credit risk:

Section	Description
	An actual or expected significant change in the operating results of the borrower. Examples include actual or expected declining revenues or margins, increasing operating risks, working capital deficiencies, decreasing asset quality, increased balance sheet leverage, liquidity, management problems or changes in the scope of business or organizational structure (such as the discontinuance of a segment of the business) that results in a significant change in the borrower's ability to meet its debt obligations
B5.5.17 (i)	• The following non-exhaustive list of information may be relevant in assessing changes in credit risk: An actual or expected significant adverse change in the regulatory, economic, or technological environment of the borrower that results in a significant change in the borrower's ability to meet its debt obligations, such as a decline in the demand for the borrower's sales product because of a shift in technology.
B5.5.51	• An entity need not undertake an exhaustive search for information but shall consider ll reasonable and supportable information that is available without undue cost or effort and that is relevant to the estimate of expected credit losses, including the effect of expected prepayments. The information used shall include factors that are specific to the borrower, general economic conditions and an assessment of both the current as well as the forecast direction of conditions at the reporting date. An entity may use various sources of data, that may be both internal (entity-specific) and external. Possible data sources include internal historical credit loss experience, internal ratings, credit loss experience of other entities and external ratings, reports and statistics. Entities that have no, or insufficient, sources of entity-specific data may use peer group experience for the comparable financial instrument (or groups of financial instruments).

3. Forward-looking Elements in CRMS Risk Rating to derive Probability of Default (PD)

No	Category	Forward-looking Element	MFRS 9 Compliant
1.	Financial (60%)		
	1. Historical	NA	NA
	2. Projection	• The obligor's ability of generating cash and covering the short-term obligations and interest in the future is evaluated. • The CRMS system allows financial forecasts of up to three future years and provides the ability for users to add projected scenarios. • The assumptions on the projection would be guided based on the overall economic outlook gathered by our Advisory Research Department ("ARD") and / or changes in business environment derived from Obligor's perspective backed by independent industry research reports. • The Future Assumptions section expressed the list of assumptions used to forecast the Future Balance Sheet and Future Income Statement is composed of the following main assumptions: a) Sales Assumptions b) Cost of Sales (COS) Assumptions c) Selling, General and Administration (SG&A) Assumptions d) Dividends and Injections Assumptions e) Taxes/Zakat Assumptions f) Minority Interest Assumptions g) Receivables Assumptions h) Inventory Assumptions i) Payables Assumptions j) Other Working Capital Accounts Assumptions k) Related Parties Assumptions l) Fixed Assets Assumptions m) Intangible & Goodwill Assumptions	B5.5.1 : Hence, when reasonable and supportable information that is more forward-looking than past due information is available, it must be used to assess changes in credit risk. B5.5.17 (g) : The following non-exhaustive list of information may be relevant in assessing changes in credit risk: An actual or expected significant change in the operating results of the borrower. Examples include actual or expected declining revenues or margins, increasing operating risks, working capital deficiencies, decreasing asset quality, increased balance sheet leverage, liquidity, management problems or changes in the scope of business or organizational structure (such as the discontinuance of a segment of the business) that results in a significant change in the borrower's ability to meet its debt obligations. B5.5.51 : An entity need not undertake an exhaustive search for information but

No	Category	Forward-looking Element	MFRS 9 Compliant
		n) Bank Debt Assumptions o) End of Service Provisions Assumptions (refer to Appendix 3) • Additionally, the projections would be put through a Sensitivity Analysis which determines the vulnerability to changes in Obligor's future cash generating ability. • Projections represent future events that are expected to take place. Although projections are invariably inaccurate in absolute terms, they are expected to provide a picture of the obligor's future cash generating ability. To counter-act any bias introduced by users to enhance future performance, projections are balanced by Sensitivity Analysis, which measures volatility and sensitivity (addressed later). • The system automatically forecasts three future years using historical trend analysis (including volatility and CAGR) but provides users with the ability to add and use their own forecast scenarios. • As with the historical financials, the weights are distributed around the future success of the obligor's business model, carrying the same weight of the Financial Risk Rating: a) Expected future ability to generate cash in the future b) Expected future ability to cover future short-term obligations and interest c) Expected future reliance on bank debt (leverage)	shall consider all reasonable and supportable information that is available without undue cost or effort and that is relevant to the estimate of expected credit losses, including the effect of expected prepayments
	3. Sensitivity	• It highlights the impact of changes in the major assumptions on the obligor's projected NOCF. The major assumptions are as follows: a) Sales b) Cost of Goods Sold c) Selling, General and Administrative d) Receivables Aging	B5.5.1 : Hence, when reasonable and supportable information that is more forward-looking than past due information is available, it must be used to assess changes in credit risk.

No	Category	Forward-looking Element	MFRS 9 Compliant
		e) Inventory Aging f) Payable Aging	
2.	Industry (15%)		
	1. Source of information	• NA	NA
	2. Industry specific	• The obligor's performance within the industry will be evaluated thoroughly as the assessment cover on the condition of the production facilities and technology required in the production as well as its market image and share • Specific question on the regulatory, economic, or technological embedded in the qualitative part of CRMS, such as: 1. Does the company invest in R&D to improve its product/services & processes? 2. Does the company benefit from any rights or patents on the technologies it uses?	B5.5.17 (i) : The following non-exhaustive list of information may be relevant in assessing changes in credit risk: An actual or expected significant adverse change in the regulatory, economic, or technological environment of the borrower that results in a significant change in the borrower's ability to meet its debt obligations, such as a decline in the demand for the borrower's sales product because of a shift in technology.
	3. Company Specific	• The obligor's performance within the industry will be evaluated thoroughly as the assessment cover on the condition of the production facilities and technology required in the production as well as its market image and share	B5.5.51 : An entity need not undertake an exhaustive search for information but shall consider all reasonable and supportable information that is available without undue cost or effort and that is relevant to the estimate of expected credit losses, including the effect of expected prepayments
		• Specific question on the type of product, characteristics of the financial instruments and the borrower as well as the geographical region embedded in the qualitative part of CRMS, such as: • Does the company provide a range of products/services within each product/service type?	

No	Category	Forward-looking Element	MFRS 9 Compliant
	4. Technology	• Specific question on the regulatory, economic, or technological embedded in the qualitative part of CRMS, such as: 1. Does this business require technology? 2. Does the company invest in R&D to improve its product/services & processes? 3. Does the company benefit from any rights or patents on the technologies it uses?	B5.5.17 (f) : The following non-exhaustive list of information may be relevant in assessing changes in credit risk: Existing or forecast adverse changes in business, financial or economic conditions that are expected to cause a significant change in the borrower's ability to meet its debt obligations, such as an actual or expected increase in interest rates or an actual or expected significant increase in unemployment rates.
	5. Other SWOT	• NA	NA
3.	Environment (10%)	• The evaluation is made using a series of questions and answers on the Obligor's environment. • It assesses the environment in which the Obligor operates at the following three levels: a) Economic: Macro-economic environment of the Obligor's country of operation b) Political: Political stability of the Obligor's country of operation c) Contagion: Degree to which the Obligor's environment is impacted by the geopolitical environment of neighboring countries • The above are reasonable and supportable information that more forward looking than historical financial spreading.	B5.5.1 : Hence, when reasonable and supportable information that is more forward-looking than past due information is available, it must be used to assess changes in credit risk. B5.5.51 : An entity need not undertake an exhaustive search for information but shall consider all reasonable and supportable information that is available without undue cost or effort and that is relevant to the estimate of expected credit losses, including the effect of expected prepayments.
	1. Economic	• Specific question on the regulatory, economic, or technological embedded in the qualitative part of CRMS, such as: 1. How much is the real GDP growth rate in the country of operation?	B5.5.17 (f) : The following non-exhaustive list of information may be relevant in assessing changes in credit risk:

No	Category	Forward-looking Element	MFRS 9 Compliant
		2. How much is the level of inflation in the country of operation? 3. What best describes the current state of the economy? 4. What is the unemployment rate in the company's country of operation? 5. Are the country's economic sectors well diversified or are they concentrated in 1-2 major sectors? 6. Is the market subject to local currency devaluations?	Existing or forecast adverse changes in business, financial or economic conditions that are expected to cause a significant change in the borrower's ability to meet its debt obligations, such as an actual or expected increase in interest rates or an actual or expected significant increase in unemployment rates. B5.5.51 : An entity need not undertake an exhaustive search for information but shall consider all reasonable and supportable information that is available without undue cost or effort and that is relevant to the estimate of expected credit losses, including the effect of expected prepayments.
	2. Political	• Specific question on the regulatory, economic, or technological embedded in the qualitative part of CRMS, such as: 1. Is the country of operation currently facing any political turmoil/unrest? 2. Is the country of operation exposed to any war threats or terror attacks? 3. Are there any street turmoil such as riots, demonstrations, etc?	B5.5.17 (f) : The following non-exhaustive list of information may be relevant in assessing changes in credit risk: Existing or forecast adverse changes in business, financial or economic conditions that are expected to cause a significant change in the borrower's ability to meet its debt obligations, such as an actual or expected increase in interest rates or an actual or expected significant increase in unemployment rates.

No	Category	Forward-looking Element	MFRS 9 Compliant
	3. Contagion	• Specific question on the regulatory, economic, or technological embedded in the qualitative part of CRMS, such as: 1. What is the general state of the economy faced by nearby neighboring countries? 2. Are neighboring countries exposed to any war threats or terror attacks? 3. Is there any political turmoil/unrest in the region? 4. Has the country been impacted by the unrest in neighboring countries in the past?	B5.5.17 (f) : The following non-exhaustive list of information may be relevant in assessing changes in credit risk: Existing or forecast adverse changes in business, financial or economic conditions that are expected to cause a significant change in the borrower's ability to meet its debt obligations, such as an actual or expected increase in interest rates or an actual or expected significant increase in unemployment rates.
4.	Management (15%)		
	1. Character	NA	NA
	2. Quality / Skills	• Financial Control & Expertise – a subsection under Quality / Skills where this will be assessed the degree of which the fiscal condition and operation is governed. • Specific question on the type of product, characteristics of the financial instruments and the borrower as well as the geographical region embedded in the qualitative part of CRMS, such as: 1. How often are budgets set and monitored? 2. Are the company's financial statements audited?	NA
	3. Organisation Structure	NA	NA

No	Category	Forward-looking Element	MFRS 9 Compliant
	4. Risk Management	• Under Operational Risk, specific question on the management problems or changes in the scope of business or organizational structure embedded in the qualitative part of CRMS, such as: 1. Does management update its technologies vis-a-vis changes in the industry's technological environment? 2. Is management aware of operational risk? 3. Does management have controls in place to manage its operational risks? 4. To what extent does management manage its operational workflow to minimize the costs of poor scheduling? (operations management? 5. Does management run any quality control or ISO processes?	B5.5.17 (g) : The following non-exhaustive list of information may be relevant in assessing changes in credit risk: An actual or expected significant change in the operating results of the borrower. Examples include actual or expected declining revenues or margins, increasing operating risks, working capital deficiencies, decreasing asset quality, increased balance sheet leverage, liquidity, management problems or changes in the scope of business or organizational structure (such as the discontinuance of a segment of the business) that results in a significant change in the borrower's ability to meet its debt obligations
	5. Regulations	NA	NA
	6. Relationship with other banks	NA	NA

- Kindly refer to page 5-13 of CRMS Risk Rating Methodology for explanations on how the respective components addresses the forward looking element mapped against each of the MFRS9 requirement

4. ECL Computation

- Stage 1

NPV EAR x PD x WALGD

- For the positive number, were
  a) EAR is the Exposure at Risk for both Authorized and Outstanding.
     - For Authorized EAR = CCF x Limits
     - For outstanding, EAR = CCF x Utilized Amount + CCF for Unutilized x Unutilized amounts
     - The EAR calculation is NPVED by dividing it with (1+EIR%). Whereby, EIR is Effective Interest Rate, interest charged plus fees
  b) PD is the month 12 probability of Default from the Loss Norm Table.
  c) The LGD. We demoed the formula for capturing the LGD portion of the exposure in our last meeting with Board members. The weighted average LGD is calculated using both clean and collateralized exposure, the latter associated with an LGD as follows:

     Weighted Average LGD = [(Clean Portion x Loss Norm LGD) + (Secured Portion x Collateral LGD x (1-Haircut%-Volatility%-Cost of Recovery%-FX differential%) / (1+Effective Interest Rate)^Expected period of Liquidation)] / Total EAR

  d) WALGD is the Weighted Average Loss Given Default that is applicable to the lower value of the LGD is associated with the Type of Facility (e.g., Structured Finance), Collateral Security (as dictated by Central Banks), or as per the Loss Norm Table. To calculate the weighted average, the following steps are used:
     - First the system NPVs the collateral current value x (1-Haircut% - Volatility% - Cost of Recovery% - FX Differential %)/(1+EIR%) ^EPL where
       - EPL is the expected period of liquidation, the average number of years the Bank takes to convert the collateral into cash (usually due to litigation proceedings)
       - Haircut is that applied by the Central Bank for that type of collateral
       - Volatility is that experienced for that collateral if not captured in the Haircut
       - Cost of Recovery is the average cost of exercising the Bank's right under the collateral
       - FX differential is the volatility of the underlying currency in case it is different than that for the exposure
     - The system then splits the NPV EAR into two portions:
       - The secured portion which carries the collateral LGD
       - The unsecured portion which carried the default LGD from the Loss Norm Table (normally 45%)

- The weighted average LGD is the calculated as:

WALGD = (Secured x Collateral LGD + Unsecured x Loss Norm LGD) / NPV EAR

e) The resultant ECL is pro-rated to the remaining period in case the facility maturity is less than one year.

- There are several forward-looking elements in determining the LGD per collateral. Even the use Basel's 45%, the haircuts etc. are forward looking elements.
- Facility Structuring and Share of Wallet %. To cap things off, the client is also categorized based on the PD factors (1 above) and the SOW% is determined accordingly. This helps maintains a lid on the overall EAR in the portfolio in relations to PD and LGD, and therefore a lid over the Portfolio Risk Rating.

- Stage 2 & 3

  - For Stage 2 and 3 and where the exposure exceeds one year, the calculation is lifetime across the whole exposure horizon, and the user has the option of choosing one of two methods:

Simplified lifetime: **Σ (EAR / (1+EIR) ^n) x PD x WALGD**

Where:

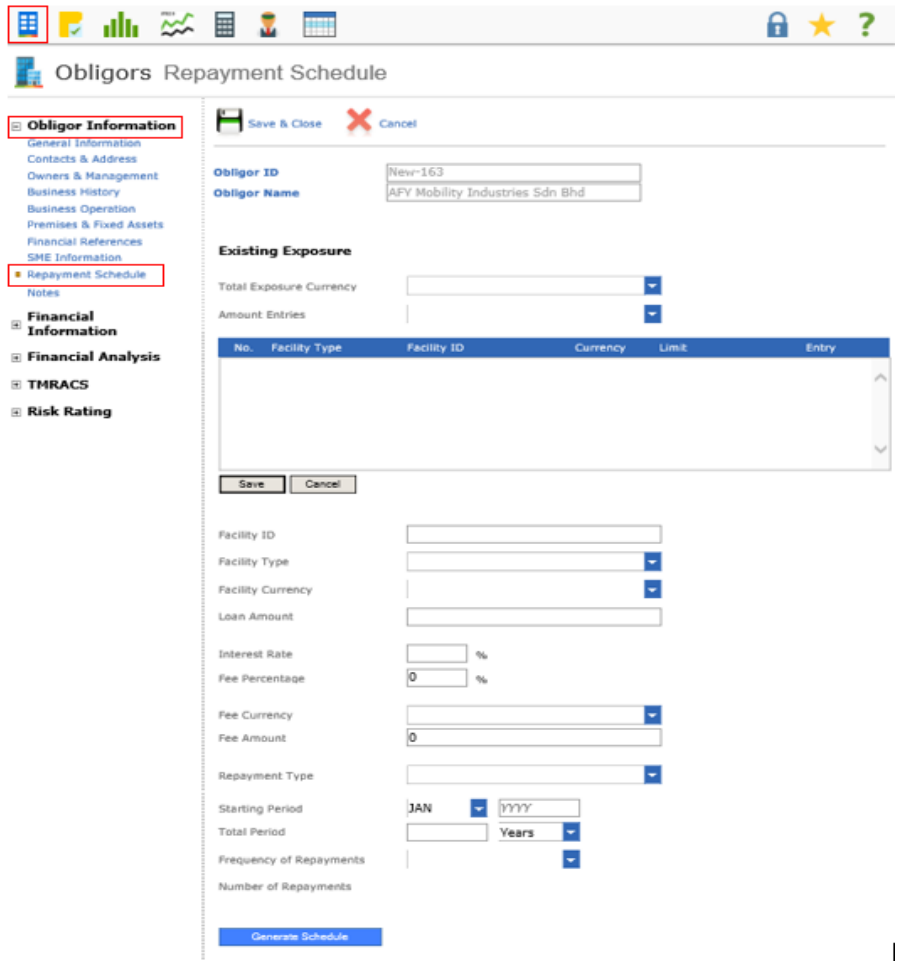
- EAR is annual values
- "n" denotes the annualized values at the nth year.
- EIR and NPV are the same as that calculated in the 12 months formula above.

Complex lifetime: **(Stage 1 NPV EAR y1 x PD y1 x WALGD) + Σ (EARn x Δ PDn x WALGD) / (1+EIR) ^n**

Where:

- "n" denotes the annualized values at the nth year
- Stage 1 portion is the same as that calculated in the 12 months formula earlier.
- EAR y1 is the first-year Exposure at Risk
- PD y1 is the 12 months PD rate that is applied to the EAR y1.
- WALGD is the same weighted average LGD outlined earlier.
- EAR n is the Exposure at Risk at year "n".
- Δ PDn is the incremental PD rate from the Transition Matrix that is applied at year "n".

5. Repayment Schedule



- Facility Repayment Schedules of all term facilities needs to be captured. This Facility Repayment Schedules of all term facilities needs to be captured. This can be done manually in the Obligor Information section of the CRMS, or through a Repayment Import routine using an excel file.

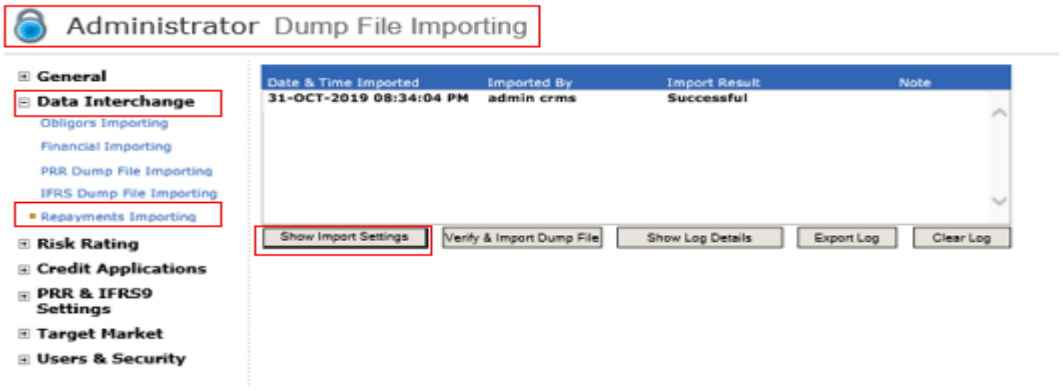
- The manual entry of a repayment schedule as follows:

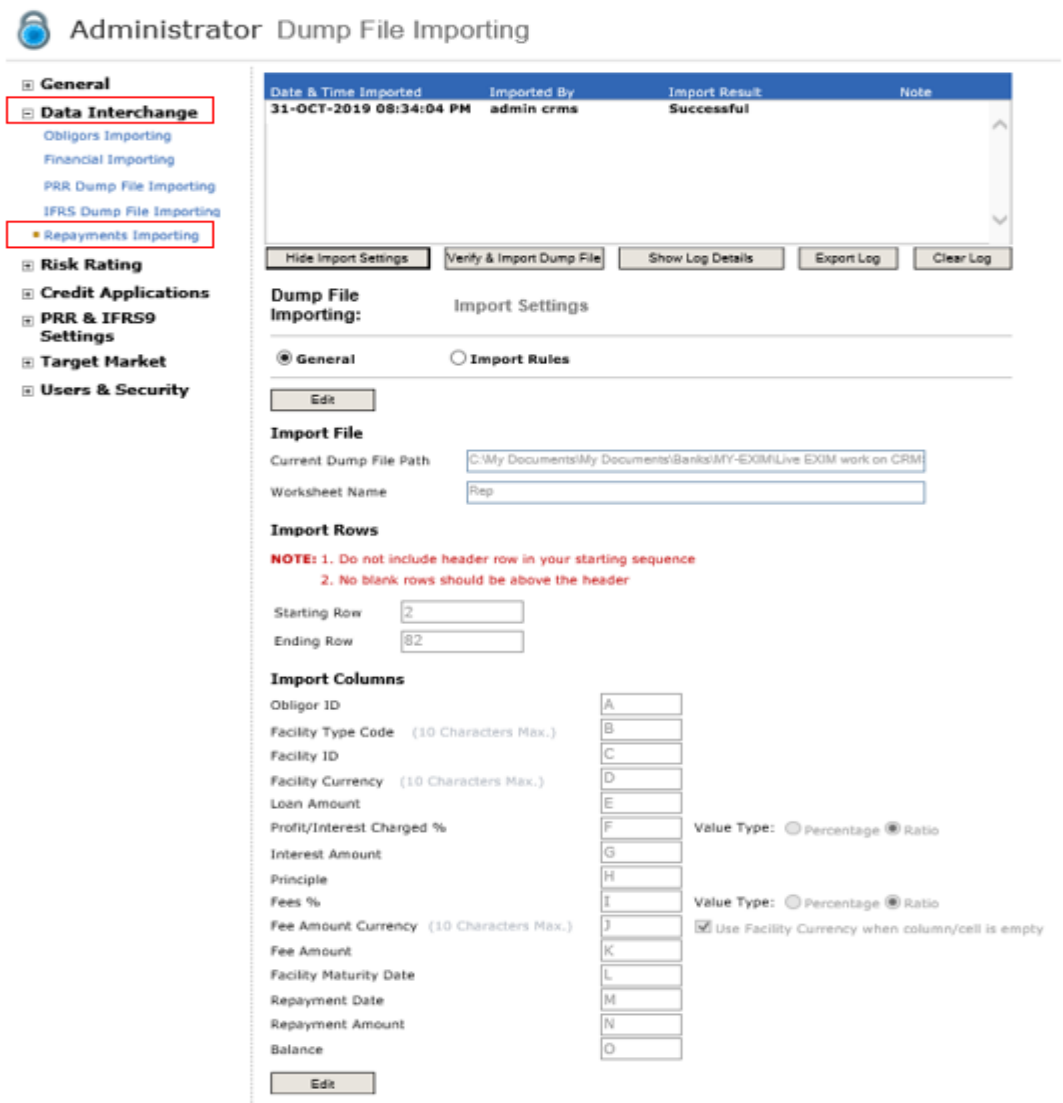
  "**Obligor**" Section → **Obligor information** → **Repayment Schedule** (Sub Section) → **Add** Facilities → Define Parameter → Press on "**Generate Schedule**" → **Save**

- For an imported Repayment Schedule:

  Repayment Import Routine:

  "**Admin**" section → **Data Interchange** → **Repayments Importing** → **Show Import Settings** → **General** → **upload** repayment schedule Dump File → set **parameters** → **save** → **verify and import** → **begin importing**


Administrator Dump File Importing
General
Data Interchange
Obligors Importing
Financial Importing
PRR Dump File Importing
IFRS Dump File Importing
Repayments Importing
Risk Rating
Credit Applications
PRR & IFRS9 Settings
Target Market
Users & Security
Date & Time Imported
Imported By
Import Result
Note
31-OCT-2019 08:34:04 PM
admin crms
Successful
Show Import Settings
Verify & Import Dump File
Show Log Details
Export Log
Clear Log


Administrator Dump File Importing
General
Data Interchange
Obligors Importing
Financial Importing
PRR Dump File Importing
IFRS Dump File Importing
Repayments Importing
Risk Rating
Credit Applications
PRR & IFRS9 Settings
Target Market
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Date & Time Imported
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Note
31-OCT-2019 08:34:04 PM
admin crms
Successful
Hide Import Settings
Verify & Import Dump File
Show Log Details
Export Log
Clear Log
Dump File Importing:
Import Settings
General
Import Rules
Edit
Import File
Current Dump File Path
C:\My Documents\My Documents\Banks\MY-EXIM\Live EXIM work on CRM\
Worksheet Name
Rep
Import Rows
NOTE: 1. Do not include header row in your starting sequence
2. No blank rows should be above the header
Starting Row
2
Ending Row
82
Import Columns
Obligor ID
A
Facility Type Code
(10 Characters Max.)
B
Facility ID
C
Facility Currency
(10 Characters Max.)
D
Loan Amount
E
Profit/Interest Charged %
F
Value Type:
Percentage
Ratio
Interest Amount
G
Principle
H
Fees %
I
Value Type:
Percentage
Ratio
Fee Amount Currency
(10 Characters Max.)
J
Use Facility Currency when column/cell is empty
Fee Amount
K
Facility Maturity Date
L
Repayment Date
M
Repayment Amount
N
Balance
O
Edit

Lifetime Term Structure

- The Transition Matrix can be found in the Admin Section under “Risk Rating” and “Transition Matrix”. The one-year Transition Matrix is to be updated at least once a year on 15th August by clicking on the Modify Button. The components for the Transition Matrix will be gather from **S&P Global rating, being one of the leading global External Rating Agency.** The system will provide automatically up to 20 years Transition Matrices and subsequently calculate the incremental Probability of Defaults used in Complex Lifetime ECL calculations. The system will also calculate the Incremental PDs used in the Complex Lifetime calculation using these transaction matrices.

- **S&P Transition Matrix** is a widely used tool in credit risk management and financial analysis. It provides a systematic way to estimate the probability of credit rating transitions, which is essential for assessing credit risk and making informed investment decisions. Below are some reasons why the S&P transition matrix is important:

    i. **Provides a Standardized Approach**. The S&P transition matrix provides a standardized approach for estimating the probability of credit rating transitions, which makes it easier to compare credit risk across different issuers and sectors.

    ii. **Widely used by the Bank and Analysts**. The S&P transition matrix is widely used by the Bank and analysts in credit risk modelling and portfolio management, which means that its estimates are well-regarded and can influence investment decisions.

    iii. **Accepted by most Regulators and Credit Rating Agencies**. The S&P transition matrix is used by regulators and credit rating agencies to assess the creditworthiness of financial institutions and issuers, which can impact their ability to access funding and its funding cost.

    iv. **Provides insights into Credit Rating Trends.** The S&P transition matrix provides insights into credit rating trends over time, which can be useful for identifying systemic risks and the impact of macroeconomic factors on credit quality.

- Why S&P transition matrix is useful in finance?

    1. **Risk Assessment**: It provides a quantitative way to assess credit risk by analysing the probability of rating changes, the Bank can evaluate the potential risks associated with investing in a particular set of bonds.

    2. **Portfolio Management**: It is useful for portfolio managers who want to optimize their portfolio based on risk and return objectives. By analysing the transition probabilities, they can adjust their holdings to maximise returns while minimising risk.

3. **Stress testing:** It can be used to conduct stress tests on a portfolio or a financial institution. Analysts can assess the impact of the rating changes on the portfolio or institution by various simulation scenarios.

- Overall, the S&P transition matrix is an important tool for credit risk management and financial analysis. While it is not the only source of credit risk estimates, it is widely used and provides a valuable benchmark for assessing credit risk.

- In the absence of sufficient internal data, the bank will have to adopt the process using a recognized External Rating Agency transition of one year which is acceptable proxy in IFRS 9 i.e. 1 year's transition matrix of S&P (Refer to Appendix E for details).

- IFRS 9 financial instruments requires entities to recognize and measure financial assets based on their ECL. In this approach, entities may use historical data including transition matrices to assess the PD and other credit losses.

- Subsequently, the S&P transition matrix may be averaged with a matrix from Moody's and Fitch if needed to capture a broad spectrum of results.

- All three primary rating agencies – S&P, Moody's, and Fitch, provide excellent transition matrixes. The three matrixes provided are meaninglessly different and the choice of one over the other would have had no material impact on the outcome. While EXIM had selected S&P, we could just as easily have selected one of the others, as we don't believe that one of the three is superior. Similarly, other banking institutions across the world use one of these matrixes that would be equally hard pressed to recommend one over the other.

- Furthermore, the Markov Chain methodology can be adopted to calculate the rest of the years. This methodology calculates the following year's transition by multiplying each item in the same row with the corresponding item in the same column for that particular rating. The Markov Chain methodology is accepted by IFRS 9 as an alternative to capturing actual defaults from a portfolio over the same period (data which EXIM does not have in any case).

- Once completed, the Bank will calculate the incremental rates (difference between the years) for only the default state (column D) and apply that in the life-time calculation of the PD (discounted to present). The increments for years 2 onwards are multiplied by the expected exposure (in the repayment schedule) for each of the years, added up and then discounted to present.

  The summation of the values in Transition Matrix shall be 100%. Any changes to the values shall require CRO's approval.

- The parameters are set in the Admin section. The step to set the parameters for Transition Matrix is as below:

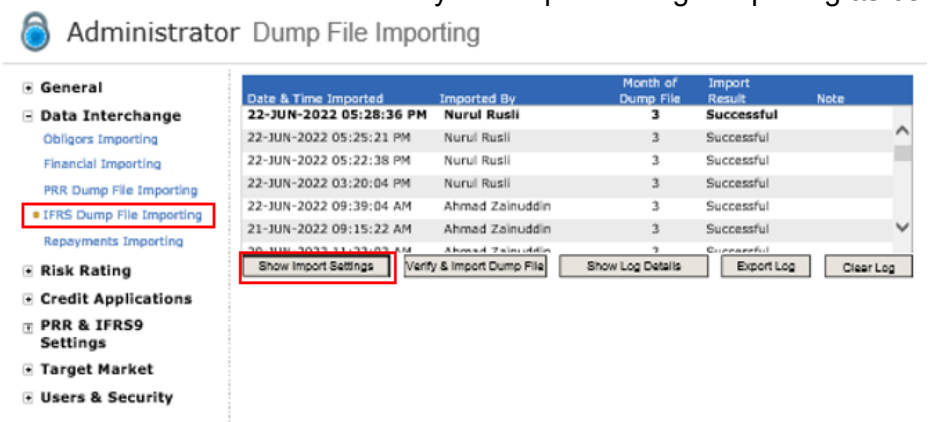
Log in to the CRMS account

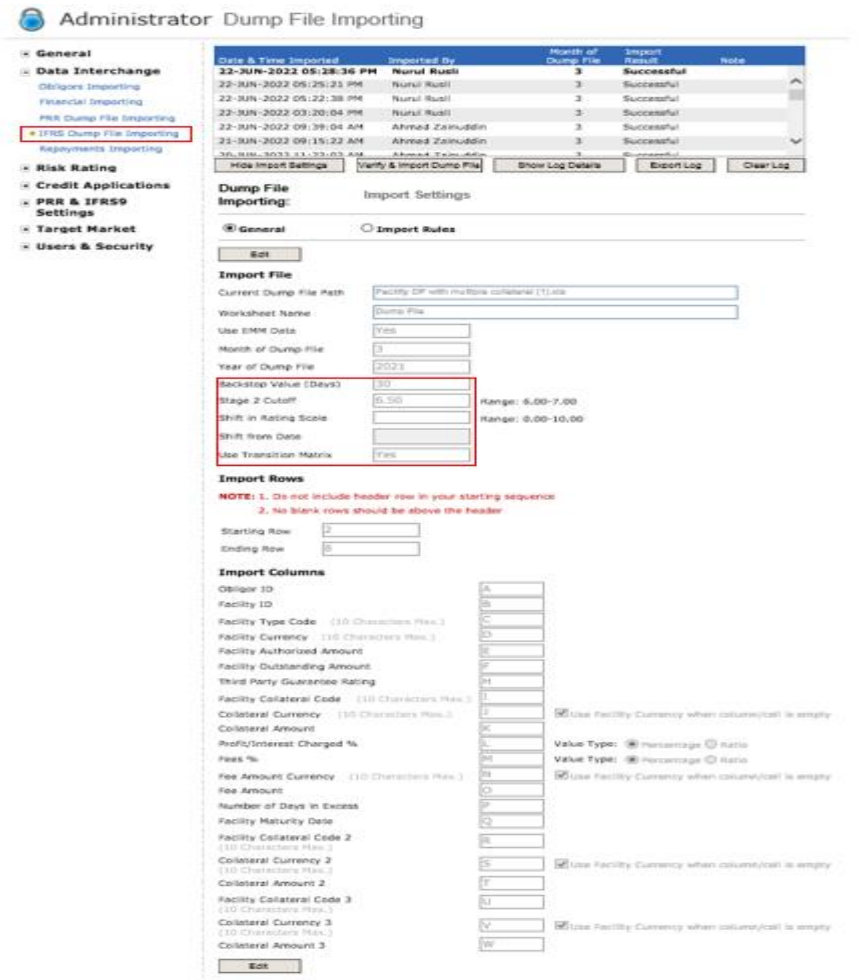


Click on System Administrator



- IFRS Dump File Importing → Show Import Settings → set parameters “Back Stop Value”, “Stage 2 Cutoff”, “Shift in Rating Scale”, “Shift from Date” and “Use of Transition Matrix” → save → verify and import → begin importing as below:





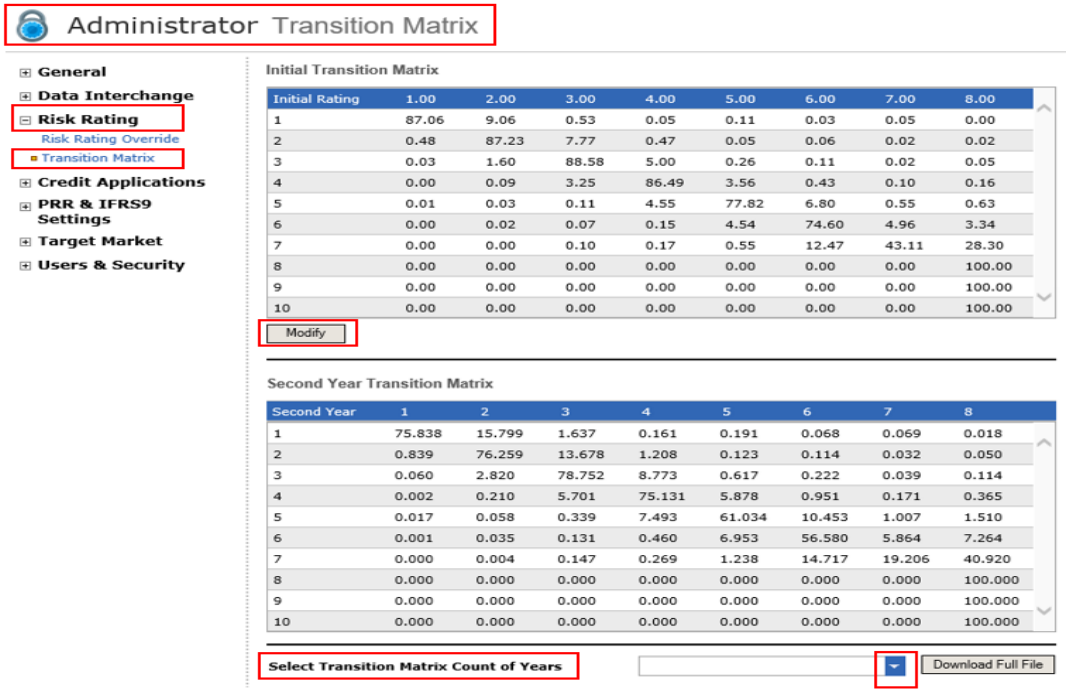
- Note: To follow the sequence in CRMS for importing the Dump File.

- Furthermore, the step for updating the one-year Transition Matrix are as below:

  Click on System Administrator



Risk Rating → Transition Matrix → Modify (Update the one-year Transition Matrix) → save. To update the Transition Matrix, refer to the Transition Matrix from S&P Global Rating.

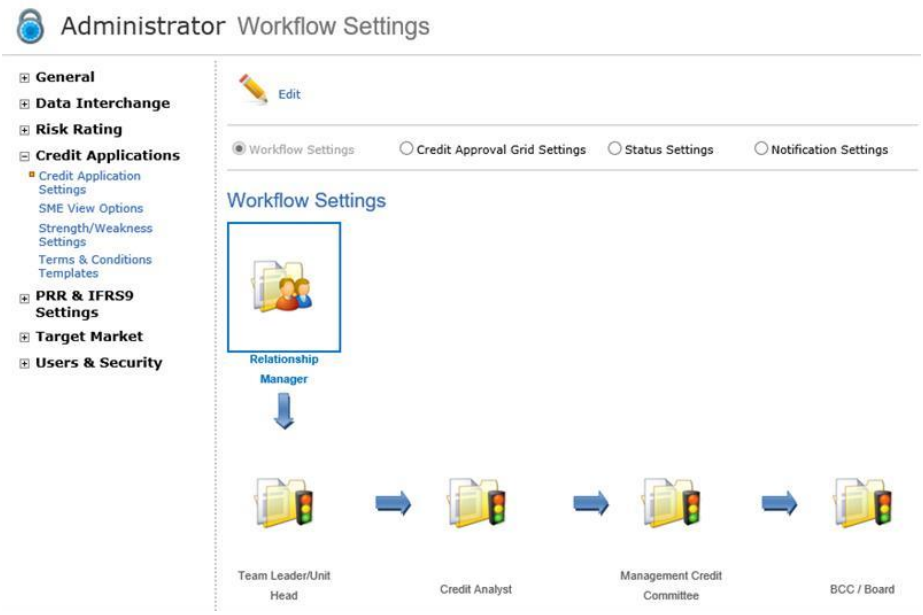


6. Management Overlay
   i. Rating overrides – The system allows the Bank to use TMRAC to manage exposure within the portfolio. If the Bank believes that the rating of Obligors are unduly high or low, then it has the option to override the ratings. The overridden ratings (if used) will be the ones that impact the ECL calculations.

CHAPTER 4: REPORTING PROCESS FLOW

1. Credit Application Module

ii. Credit application consists of 5 credit approval process, as portrayed in picture below:



iii. The Relationship Manager (RM) will prepare the credit proposal.

iv. Once the RM has completed all the thirteen (13) sub sections under Credit Proposal section, then it will flow to the next approval process which is Team Lead.

v. Team lead will review all the information key in by the RM before the level of credit risk that the borrower/prospect borrower in the proposal being assessed by the Credit Analyst.

vi. The Credit Proposal will then flow to the Management Credit Committee (MCC) approval before the final stage of approval which is BCC/Board.

vii. The approval level of the Credit Proposal will depends on the loan amount as per ruled out in the Approving Authorities and Authority Limits.

2. The MFRS9 Report

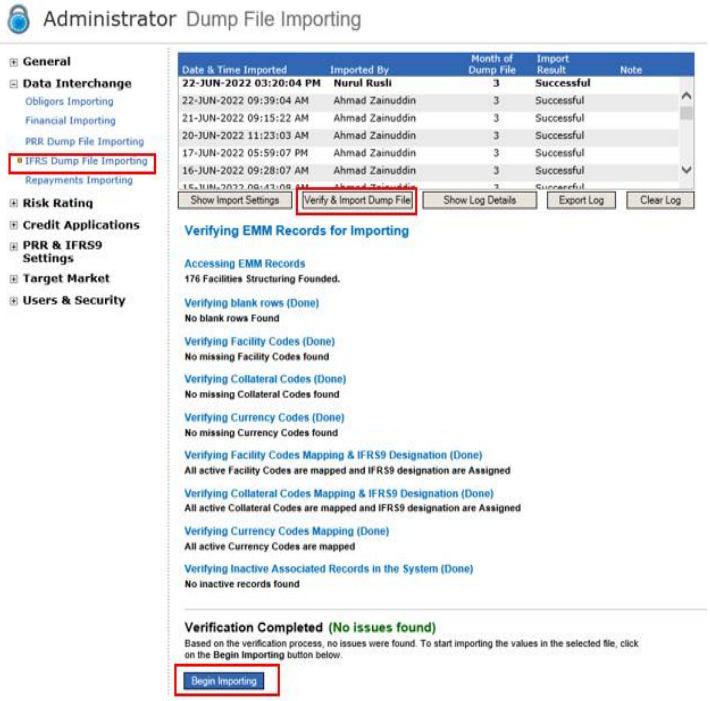
viii. Log in to the CRMS account:



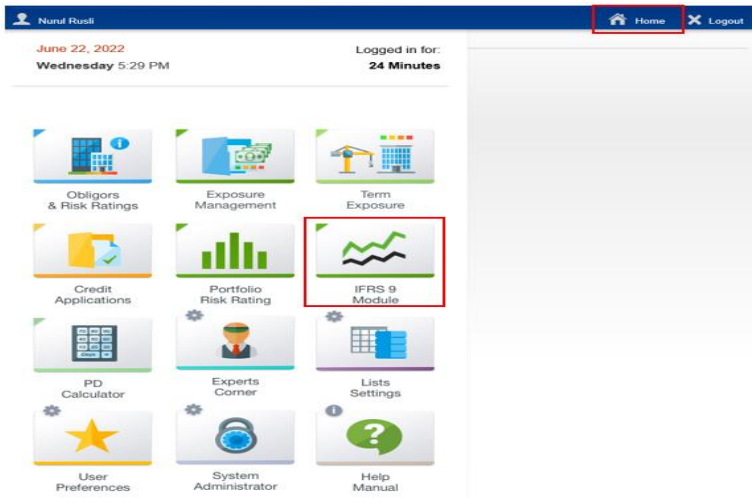
ix. Click on System Administrator:



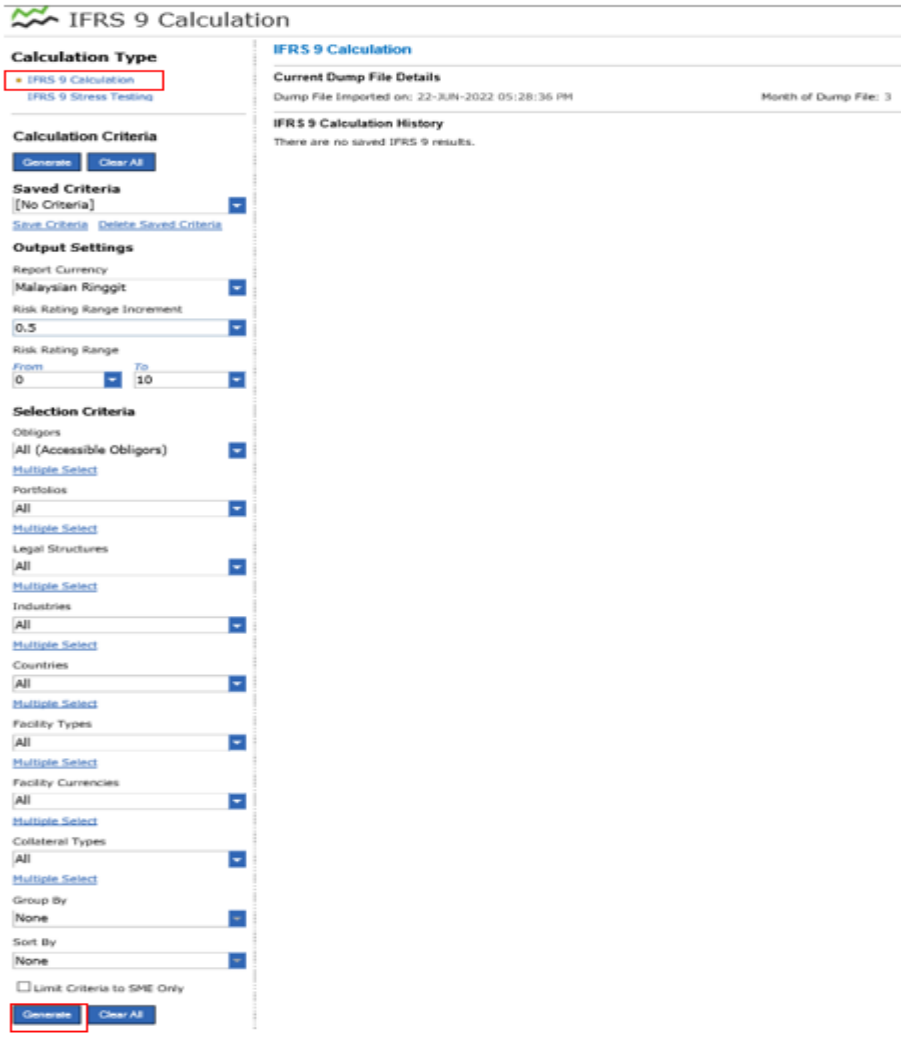
x. Then click on IFRS Dump File Importing → Verify & Import Dump File → Begin Importing



xi. Once the importing process was completed, click Home → IFRS 9 Module



xii. Click on IFRS 9 Calculation → Generate (applicable for all customers). Select the criteria listed on Calculation Criteria if for specific criteria i.e., Risk Rating Range, SME or Report Currency.



xiii. The report generated as below. Click on “Print Report” or “Export to Excel” for the record.

IFRS

Calculation Results

Dump File Imported on: 22-JUN-2022 05:28:36 PM **Month of Dump File:** 3

Generated On: 22-JUN-2022 06:04:48 PM **Generated By:** Nurul Husna Rusli

Report Currency: Malaysian Ringgit

Report Criteria:

Risk Rating Range :	0 to 10 (Increment of 0.5)
Obligors:	All (Accessible Obligors)
Portfolios:	All
Legal Structure:	All
Industries:	All
Countries:	All
Facility Type:	All
Facility Currency:	All
Collateral Type:	All
Group By:	None
Sort By:	None
Criteria Limited to SMEs:	No
Criteria Output:	Show Included Obligors

Save Results

Obligor & Facility Risk Rating Frequency Distribution as of 22-JUN-2022 06:04:24 PM

LOW RISK RATING RANGE	HIGH RISK RATING RANGE	TOTAL NUMBER OF OBLIGORS IN RANGE ORR	TOTAL NUMBER FACILITIES IN RANGE FRR
0.00	0.50	0	0
0.51	1.00	0	0
1.01	1.50	0	0
1.51	2.00	0	0
2.01	2.50	0	0
2.51	3.00	0	0
3.01	3.50	0	0
3.51	4.00	1	1
4.01	4.50	1	0
4.51	5.00	4	5
5.01	5.50	5	2
5.51	6.00	9	6
6.01	6.50	13	28
6.51	7.00	11	11
7.01	7.50	9	15
7.51	8.00	2	4
8.01	8.50	0	0
8.51	9.00	0	21
9.01	9.50	0	0
9.51	10.00	58	78

Generate Chart

IFRS 9 Calculation Results Summary

RESULTS SUMMARY	AUTHORIZED	OUTSTANDING
Total Facility Expected Loss	2,658,217,662	1,265,874,397
Total Facility Expected Loss Stage 1	17,225,822	15
Total Facility Expected Loss Stage 2	324,299,065	42
Total Facility Expected Loss Stage 3	2,316,692,775	1,265,874,340
Total Facility Unexpected Loss	3,358,329,408	1,653,004,124
Total Collective Obligor Exposure At Risk	9,945,394,542	9,890,766,227
Total Collective Obligor Expected Loss	2,534,187,091	1,020,963,729
Total Amount	8,495,183,850	1,730,285,578
RAROC Total	(2,779,250,830)	(1,346,760,207)
RORAC %	(77%)	(37%)
Total Collective Collateral Values	6,678,689,390	
Number of Obligors	113	
Number of Facilities	171	

Risk Adjusted Return on Capital

Risk Adjusted Return on Capital	AUTHORIZED	OUTSTANDING
Total Revenue	364,874,441	80,955,753
Cost of Funding	(244,174,537)	(53,524,106)
Total Net Revenue	**120,699,904**	**27,441,657**
Gross Margin %	33%	34%
Operating Expenses	(40,233,301)	(9,147,219)
Net Operating Income	**80,466,602**	**18,294,438**
Net Operating Margin %	**67%**	**67%**
Expected Loss	(2,658,217,662)	(1,265,874,397)
Net Profit after Expected Loss	**(2,577,751,060)**	**(1,247,579,959)**
Tax	0	0
Net Profit after Tax	**(2,577,751,060)**	**(1,247,579,959)**
Net Profit Margin %	**(2136%)**	**(4546%)**
Unexpected Losses	(3,358,329,498)	(1,653,004,124)
RAROC	**(2,779,250,830)**	**(1,346,760,207)**
RORAC %	(77%)	(37%)
Standardized ROC %	(215.99%)	(105.11%)

Authorized Concentration Calculation

NO.	OBLIGOR ID	OBLIGOR NAME	ORR	TOTAL EXPOSURE AT RISK	TOTAL EXPOSURE AT RISK %	CUMULATIVE TEAR %
1	EXIM000104	The Ministry of Finance Government of Lao PDR	7.20	1,689,330,004	17.0%	17.0%
2	EXIM000389	Yinson Boronia Production B.V.	6.45	858,897,708	8.6%	25.6%
3	EXIM000145	OM Materials (Sarawak) Sdn Bhd	5.27	786,472,665	7.9%	33.5%
4	EXIM000335	SKS Waterfront Pty Ltd	6.66	570,721,153	5.7%	39.3%
5	EXIM000140	Pertama Ferroalloys Sdn. Bhd.	7.33	538,434,467	5.4%	44.7%
6	EXIM000257	PT Sriwijaya Air	10.00	340,878,679	3.4%	48.1%
7	EXIM000101	Marlemore Energy (Labuan) Limited	10.00	332,819,820	3.3%	51.5%
8	EXIM000167	Padiberas Nasional Berhad	5.30	287,935,502	2.9%	54.4%
9	EXIM000171	Armada Kraken Pte. Ltd.	4.99	279,911,254	2.8%	57.2%
10	EXIM000209	S P Setia Berhad	5.48	260,863,286	2.6%	59.8%
11	EXIM000232	Smart Glove Corporation Sdn Bhd	7.08	248,095,910	2.5%	62.3%
12	EXIM000225	Amcorp Properties Berhad	5.86	234,329,079	2.4%	64.6%
13	EXIM000173	Sapura THC Sdn Bhd	6.36	227,495,242	2.3%	66.9%
14	EXIM000144	Nautilus Tug & Towage Sdn Bhd	3.89	217,165,424	2.2%	69.1%
15	EXIM000313	Malaysian Phosphate Additives (Sarawak) Sdn Bhd	6.15	213,108,203	2.1%	71.3%
16	EXIM000160	HCD-PIBSB Co. Ltd	10.00	167,895,710	1.7%	72.9%
17	EXIM000180	Kota Laksamana 101 Limited	10.00	164,588,070	1.7%	74.6%
18	EXIM000245	British University Vietnam Company Ltd	6.96	160,310,410	1.6%	76.2%
19	EXIM000228	Intra Asia (Malaysia) Pte. Ltd.	10.00	142,350,252	1.4%	77.6%
20	EXIM000342	Probase Eswatini Pty Ltd	4.87	118,088,018	1.2%	78.8%
21	EXIM000059	PT Mahakarya Inti Buana	10.00	101,698,182	1.0%	79.8%
22	EXIM000210	SMH Rail Sdn Bhd	5.81	100,729,760	1.0%	80.9%
23	EXIM000211	Mewah-Oils Sdn Bhd	4.89	99,251,175	1.0%	81.9%
24	EXIM000115	Perusahaan Otomobil Nasional Sdn. Bhd	5.90	97,713,104	1.0%	82.8%

Outstanding Concentration Calculation

NO.	OBLIGOR ID	OBLIGOR NAME	ORR	TOTAL OUTSTANDING AT RISK	TOTAL OUTSTANDING AT RISK %	CUMULATIVE TOAR %
1	EXIM000104	The Ministry of Finance Government of Lao PDR	7.20	1,689,330,004	17.1%	17.1%
2	EXIM000389	Yinson Boronia Production B.V.	6.45	858,897,708	8.7%	25.8%
3	EXIM000145	OM Materials (Sarawak) Sdn Bhd	5.27	786,472,665	8.0%	33.7%
4	EXIM000335	SKS Waterfront Pty Ltd	6.66	570,721,153	5.8%	39.5%
5	EXIM000140	Pertama Ferroalloys Sdn. Bhd.	7.33	538,434,467	5.4%	44.9%
6	EXIM000257	PT Sriwijaya Air	10.00	340,878,679	3.4%	48.4%
7	EXIM000101	Marlemore Energy (Labuan) Limited	10.00	332,819,820	3.4%	51.7%
8	EXIM000167	Padiberas Nasional Berhad	5.30	287,935,502	2.9%	54.7%
9	EXIM000171	Armada Kraken Pte. Ltd.	4.99	279,911,254	2.8%	57.5%
10	EXIM000209	S P Setia Berhad	5.48	260,863,286	2.6%	60.1%
11	EXIM000232	Smart Glove Corporation Sdn Bhd	7.08	248,095,910	2.5%	62.6%
12	EXIM000225	Amcorp Properties Berhad	5.86	234,329,079	2.4%	65.0%
13	EXIM000173	Sapura THC Sdn Bhd	6.36	227,495,242	2.3%	67.3%
14	EXIM000144	Nautilus Tug & Towage Sdn Bhd	3.89	217,165,424	2.2%	69.5%
15	EXIM000313	Malaysian Phosphate Additives (Sarawak) Sdn Bhd	6.15	213,108,203	2.2%	71.6%
16	EXIM000160	HCD-PIBSB Co. Ltd	10.00	167,895,710	1.7%	73.3%
17	EXIM000180	Kota Laksamana 101 Limited	10.00	164,588,070	1.7%	75.0%
18	EXIM000245	British University Vietnam Company Ltd	6.96	160,310,410	1.6%	76.6%
19	EXIM000228	Intra Asia (Malaysia) Pte. Ltd.	10.00	142,350,252	1.4%	78.1%
20	EXIM000342	Probase Eswatini Pty Ltd	4.87	118,088,018	1.2%	79.3%
21	EXIM000059	PT Mahakarya Inti Buana	10.00	101,698,182	1.0%	80.3%
22	EXIM000210	SMH Rail Sdn Bhd	5.81	100,729,760	1.0%	81.3%
23	EXIM000211	Mewah-Oils Sdn Bhd	4.89	94,419,790	1.0%	82.3%

NO.	OBLIGOR ID	OBLIGOR NAME					CURRENCY
111	EXIM000317	Bio Eneco Sdn Bhd					Malaysian Ringgit
FACILITY NO.	FACILITY TYPE	FACILITY AMOUNT AUTHORIZED	FACILITY AMOUNT OUTSTANDING	Collateral	FRR	IFRS 9 EXPECTED LOSS AUTHORIZED	IFRS 9 EXPECTED LOSS OUTSTANDING
1	Supplier Financing - i	7,000,000	0	3,200,000	9.00	4,253,714	1
2	Forward Foreign Exchange Islamic	1,000,000	0	0	7.96	217,875	1
INDIVIDUAL TOTAL						4,471,589	2
COLLECTIVE OBLIGOR		8,000,000	0	3,200,000	ORR: 7.96	0	0

NO.	OBLIGOR ID	OBLIGOR NAME					CURRENCY
112	EXIM000389	Yinson Boronia Production B.V.					Malaysian Ringgit
FACILITY NO.	FACILITY TYPE	FACILITY AMOUNT AUTHORIZED	FACILITY AMOUNT OUTSTANDING	Collateral	FRR	IFRS 9 EXPECTED LOSS AUTHORIZED	IFRS 9 EXPECTED LOSS OUTSTANDING
1	Overseas Project Financing	881,057,269	0	0	6.50	25,509,262	1
INDIVIDUAL TOTAL						25,509,262	1
COLLECTIVE OBLIGOR		881,057,269	0	0	ORR: 6.45	46,544,182	0

NO.	OBLIGOR ID	OBLIGOR NAME					CURRENCY
113	EXIM000104	The Ministry of Finance Government of Lao PDR					Malaysian Ringgit
FACILITY NO.	FACILITY TYPE	FACILITY AMOUNT AUTHORIZED	FACILITY AMOUNT OUTSTANDING	Collateral	FRR	IFRS 9 EXPECTED LOSS AUTHORIZED	IFRS 9 EXPECTED LOSS OUTSTANDING
1	Buyer Credit	286,343,612	0	0	7.20	43,340,576	1
2	Buyer Credit	308,370,044	0	0	7.20	83,322,048	1
INDIVIDUAL TOTAL						126,662,624	2
COLLECTIVE OBLIGOR		594,713,656	0	0	ORR: 7.20	276,542,308	0

Print Report | Export to Excel

Save Results

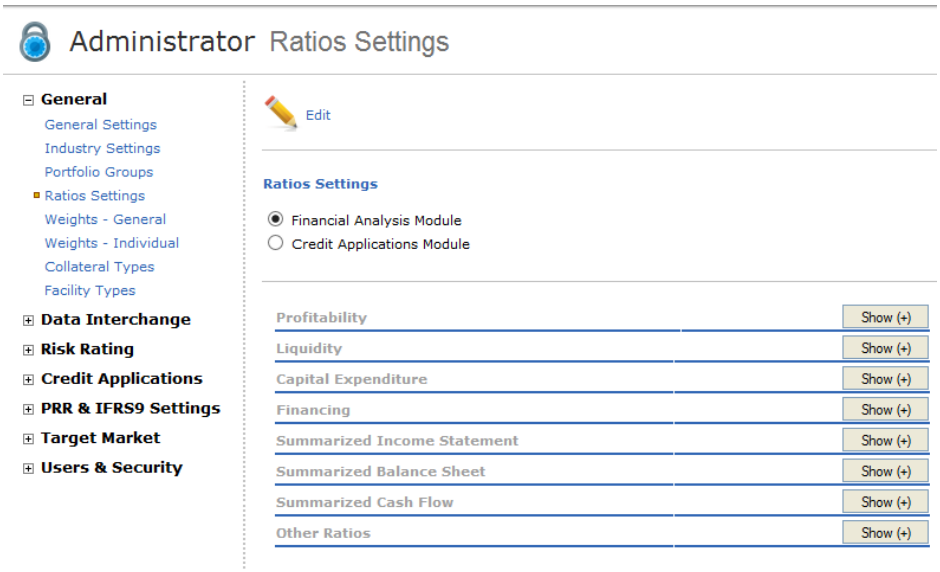
CHAPTER 5: SYSTEM MAINTENANCE

1. Administration
   - Function
     - This feature designated to the System Administrator to controls the privileges of the users in CRMS
     - The admin section divided into many sections that give the flexibility to the designated Administrator in pre-set in CRMS detailed as follows:
       - **General Settings**

       - This setting section allow the Administrator detailed as below:

         1) To upload the Bank's logo on the CRMS system
         2) To populate the local industry codes
         3) To create a new portfolio group
         4) Assigning a user to a different portfolio group
         5) To allocate the weight and parameter for Financial Analysis Module Ratio Settings and Credit Application Module Ratio Settings
         6) To set the status, add and modify the collateral types
         7) To set the status, add and modify the facility types

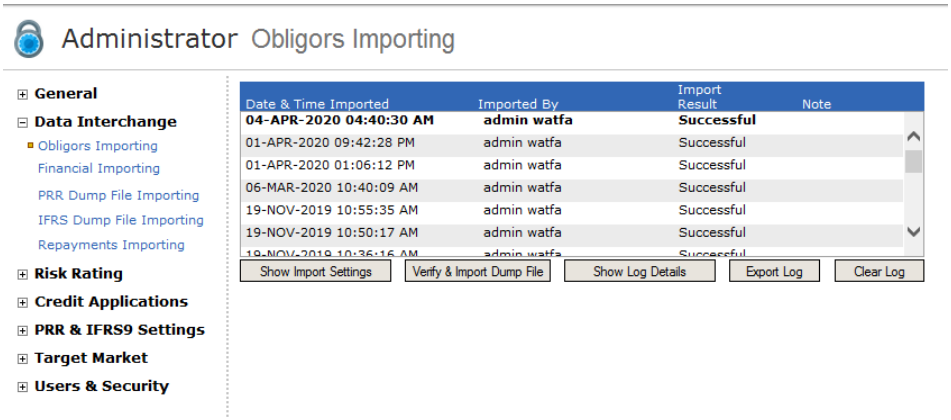
       - The snapshot of this section



       - **Data Interchange**

       - Data Interchange sections give the flexibility to the Administrator to import the obligor information, financial statements, PRR information, IFRS 9 Information and repayment schedule into CRMS through an excel Dump File with the necessary template

       - The snapshot of this section



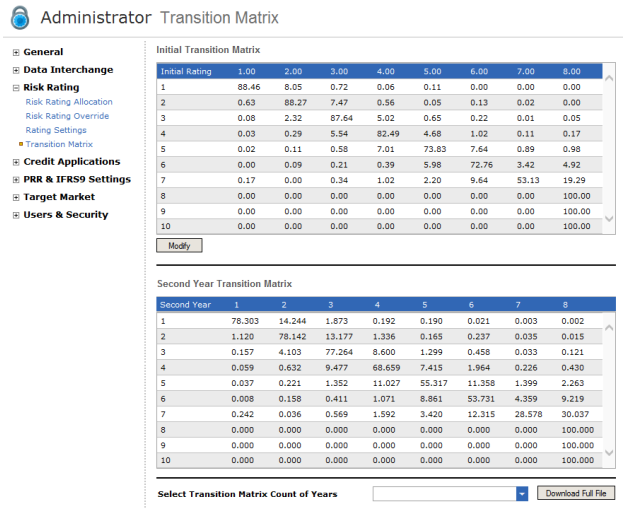
- Risk Rating

The risk rating section composed the followings:

No.	System Administrator – Risk Rating	Descriptions
1.	Risk Rating Allocations	To check the last modification done on which risk rating was chosen or a detailed log with all modification done
2.	Risk Rating Overrides	To set the rules of risk rating overrides either system risk rating or TMRACs Rating Override
3.	Rating Settings	To specify the risk rating for obligors that do not provide Financial Statements
4.	Transition Matrix	Provides the option for the Complex calculation by using incremental PDs for lifetime calculation

Table 55: Risk Rating Compositions

The snapshot of this section



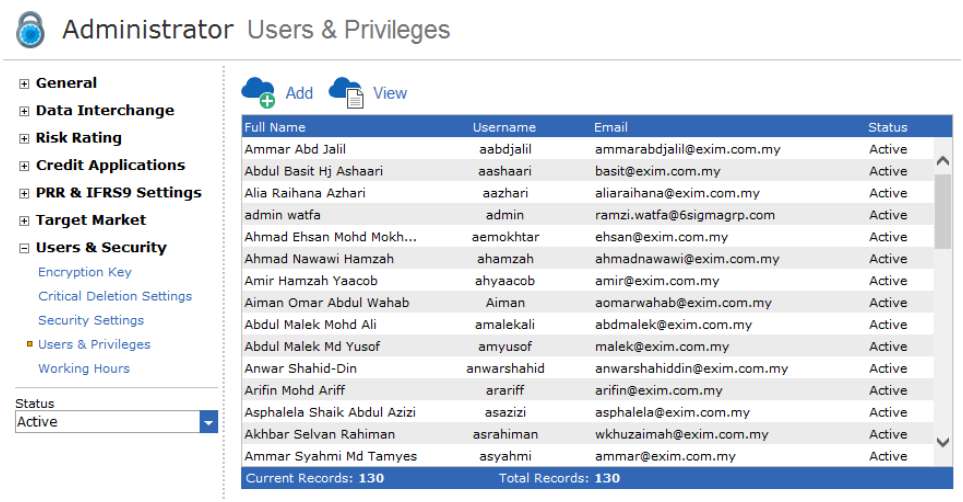
- **Users & Security Settings**

- The security sections allows the Administrator to define the number of times a user is required to enter his/her password wrong the user's account is locked out
- At this stage, the Administrator also given the flexibility to set the following privileges for all the users of the CRMS:

No.	Privileges Compositions	Descriptions
1.	General	Provides information about the User that divided into account security, active vs inactive and password sub sections
2.	Login Hours	Defines the hours of each day in which the CRMS user is authorized to enter the system
3.	Privileges	To set the privilege for obligor information and list settings modules
4.	Portfolio Groups	To set the CRMS user ability to add, view and modify the Expert Corner and Credit Application modules
5.	Log	Displays a day by day log activity for each user

Table 58: Privileges Setting Compositions

- The snapshot of the section



CHAPTER 6: PORTFOLIO RISK RATING (PRR)

- PRR is the weighted average of the FRR of the outstanding facilities. The PRR is used to review and manage the overall quality and trajectory of the Bank's credit portfolio. PRR also is an indicator in evaluating the riskiness level of the portfolio at any point in time either from the point view of the obligors, single counterparties, industry or even geography.

1. Portfolio Risk Rating Feature

   a) The Portfolio Risk Rating module was designed to assess the risk rating of a particular portfolio, to measure it Expected Loss and develop the strategies needed to preserve capital adequacy and meet Basel's Guidelines

   b) Allows the user to generate statistic or analysis at the portfolio level. This section is composed two sections of calculation types of analysis or reports:

No.	Portfolio Risk Rating Compositions	Descriptions
1.	Portfolio Risk Rating (PRR) Output	It combines the results of the obligor and facility ratings, along with facility data into a comprehensive and complete report producing the followings: • ORR & FRR frequency distributions • Portfolio Capital Adequacy • Historical Risk Adjusted Return on Capital (RAROC) • Projected Risk Adjusted Return on Capital (RAROC) • Authorized Concentration Calculation • Outstanding Concentration Calculation
2.	Portfolio Stress Testing	Allows user to generate the stress testing a selected portfolio for which a PRR run was made

Table 49: Portfolio Risk Rating Composition

   c) Portfolio Risk Rating Calculations

   - Under this section, PRR reports are produced at the level of various cuts within the entire portfolio. The selection of criteria composed of:
     1. Obligors
     2. Portfolio
     3. Legal Structures
     4. Industries
     5. Countries
     6. Facility Type
     7. Facility Currencies
     8. Collateral Types

   - As mentioned in Table 49, the ORR and FRR Frequency Distribution is one of the Portfolio Risk Rating (PRR) output. This PRR output is helps to identify the spread

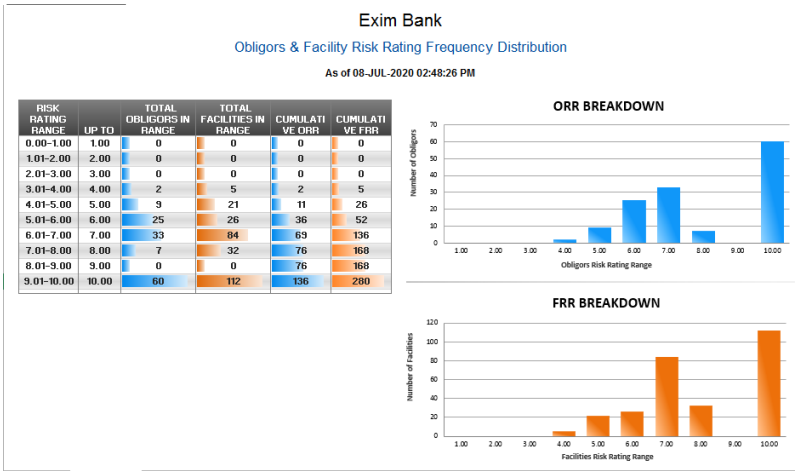
of the type of obligor by the selected portfolio. In other words, it describes how much number of obligor and facility that the Bank have at the specific risk rating range.

- This PRR output consist of the following information:
    1. Low Risk Rating Range
    2. High Risk Rating Range
    3. Total Number of Obligor in Range
    4. Total Number of Facilities in Range

- The snapshot of the ORR and FRR Frequency Distribution output

Obligor & Facility Risk Rating Frequency Distribution as of 08-JUL-2020 02:18:33 PM

LOW RISK RATING RANGE	HIGH RISK RATING RANGE	TOTAL NUMBER OF OBLIGORS IN RANGE ORR	TOTAL NUMBER FACILITIES IN RANGE FRR
0.00	1.00	0	0
1.01	2.00	0	0
2.01	3.00	0	0
3.01	4.00	2	5
4.01	5.00	9	21
5.01	6.00	25	26
6.01	7.00	33	84
7.01	8.00	7	32
8.01	9.00	0	0
9.01	10.00	60	112



RISK RATING RANGE	UP TO	TOTAL OBLIGORS IN RANGE	TOTAL FACILITIES IN RANGE	CUMULATIVE ORR	CUMULATIVE FRR
0.00-1.00	1.00	0	0	0	0
1.01-2.00	2.00	0	0	0	0
2.01-3.00	3.00	0	0	0	0
3.01-4.00	4.00	2	5	2	5
4.01-5.00	5.00	9	21	11	26
5.01-6.00	6.00	25	26	36	52
6.01-7.00	7.00	33	84	69	136
7.01-8.00	8.00	7	32	76	168
8.01-9.00	9.00	0	0	76	168
9.01-10.00	10.00	60	112	136	280

- As for Portfolio Capital Adequacy Output, this is shown the Capital Adequacy that is required to cover the Portfolio. It is calculated using both Standardized and IRB approaches which define in Table 50.

No.	Calculation Approaches	Definitions
1.	Standardized	The approach that imposed by Basel Committee and Central Bank on banks within a country
2.	Internal Rated Based (IRB)	The approach that the Bank multiplies the Exposure at Risk with the Expected and Unexpected Loss

Table 50: Portfolio Capital Adequacy Calculation Approaches

- This PRR output will generate Authorized Portfolio Capital Adequacy and Outstanding Portfolio Capital Adequacy that consists of the following information:
    1. Capital Adequacy Expected Loss
    2. Capital Adequacy Unexpected Loss
    3. Exposure & Outstanding at Risk Totals
    4. PRR Expected Loss
    5. Unexpected Loss % EAR & OAR
    6. Standardized approach Capital Adequacy
    7. Single Granulated Capital Adequacy
    8. Multiple Granulated Capital Adequacy
    9. Balance 100% RW Capital Adequacy

- The snapshot of the Portfolio Capital Adequacy Output

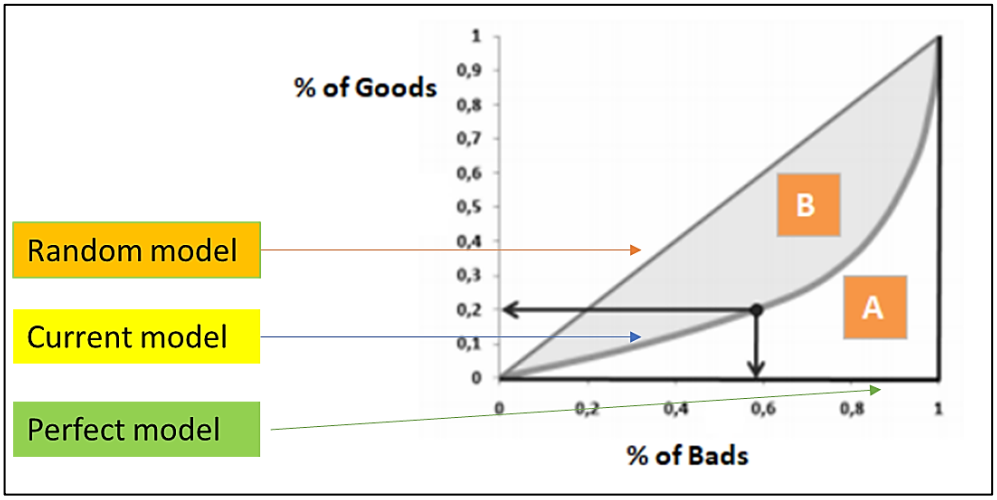
Portfolio Capital Adequacy

PORTFOLIO CAPITAL ADEQUACY	AUTHORIZED	OUTSTANDING
Capital Adequacy Expected Loss	1,242,701,333	646,289,434
Capital Adequacy Unexpected Loss	1,281,059,477	665,907,916
Exposure & Outstanding at Risk Totals	2,982,688,250	1,572,576,027
PRR Expected Loss	7.99	7.98
Unexpected Loss % EAR & OAR	43%	42%
Standardized Approach Capital Charge	220,621,872	111,372,237
Single Granulated Capital Adequacy	4,727,728	3,237,136
Multiple Granulated Capital Adequacy	0	0
Balance 100% RW Capital Adequacy	238,615,060	125,806,082

CHAPTER 7: MODEL VALIDATION

1. Validation
   - The Exim periodically review whether its credit risk measurement methodologies continue to be relevant. The assessment includes but not limited to the accuracy and discriminatory power of the credit risk measurement methodologies, whether the risk factors underlying the methodologies remain appropriate and whether the existing methodologies continue to suit the nature of the portfolio.
   - Gini Coefficient measure discriminatory power of a CRMS model. It is also known as Accuracy Ratio. It evaluates whether the probability CRMS model is able to separate good and bad customers. It is interpreted as by rejecting x% of good customers, what percentage of bad customers we reject alongside.

     Gini = B / (A+B)



   - 6 Sigma in November 2021 conducted a validation exercise using a randomly selected population of 222 of Malaysian companies as at November 2021. **The Accuracy Ratio of CRMS Financial Risk Rating was calculated at 56.19%,** which refects a good discriminatory power of the rating (very acceptable).

2. Weakness and Limitation
   - The CRMS rating engine implements a cash-flow centric model that prioritizes on the obligor's ability to generate sufficient Net Operating Cash Flow to serve its debt obligations. Consequently, obligors which requires high capital expenditure as part of their nature of business will be additionally penalized during the earlier stages of the project / development. Furthermore, obligors with strong holding of other assets than cash (e.g. property, inventory or investment) may not be able to generate a Financial Risk Rating that truly depicts the overall financial position of the obligor. The above may include but not limited to obligors that requires high stockholding, has longer asset conversion cycles, and / or obligors within capital-intensive sectors.
   - Secondly, the credit rating system maintains a "one size fits all" Financial Risk Rating weightage across all sectors / industries. The CRMS model does not recognise and place higher weightage to selected key ratios in different types of sectors accordingly.

As a result, a slower asset moving business with higher profit margin i.e. Property Developer will be assessed in the same manner as a faster asset moving business with lower profit margin i.e. Trading Company.

- Gini Coefficient uses the public data that 6 Sigma acquired to conduct the financial rating. Therefore, it does not have qualitative elements as no interview was conducted with these tested companies.
- The above limitations in the CRMS rating methodology shall be highlighted in the Financing Proposal Paper / Credit Risk Analysis Report as and when applicable.

CHAPTER 8: DATA SANITY CHECK FOR CRMS MODEL VALIDATION

8.1 Introduction

This section defines the process flow for an effective dana sanity check for the purpose of the annual CRMS model validation purpose.

8.2 Scope of Coverage

The scope of coverage covers all the related process as well as responsible parties in performing data sanity check annually.

8.3 Objective

i. The objective of data sanity check is to study the data accuracy, data completeness and overall data trend. The analysis identifies data quality issues such as illogical data values, extreme values, and invalid codes due to inaccurate data entry, or system error.

ii. Data sanity checking is a key process in the model validation exercise. RMD will ensure that the data sanity checking is conducted and documented accordingly in the annual modelling and validation exercise.

8.4 Overview

8.4.1 The following are the details of the analysis performed to ascertain the accuracy of the data used:

i. Data Completeness

- Check on missing values by observing parameters of the working file: NOCF, DSCR Total Debt, DSCR Short Term Debt, DSCR (Interest), Receivable Aging, Inventory Aging and Payable Aging.

ii. Data Accuracy

- Check on the weightage distribution for the parameters in the questionnaire with the methodology document as this variable was used to calculate the final ORR.
- Check on the formulas applied in the working file with the methodology document.
- Check on invalid codes.
- Check on extreme values (i.e. Zero or blank denominators).

8.4.2 The maker-checker concept applies the principle of one person initiating the process and the other person completes the process. Maker-checker system is put in place to mitigate the risk of errors/mistake or oversight. A maker refers to a person who prepares the document whilst the checker have to be another person.

All documents from an individual maker must be forwarded to the checker for perusal. If the document is in order and factually correct, the checker shall indicate his/her approval.

If the document is found not in order or erroneous, the checker may return the document to the maker for further review with or without amendment.

8.5 Process Flow

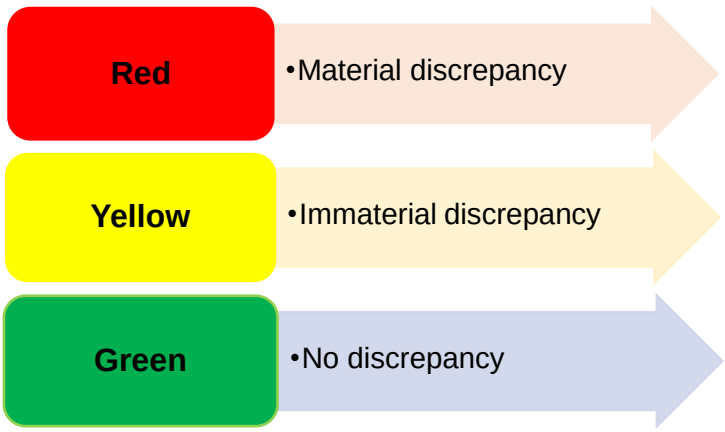
8.5.1 The data sanity checking process will be performed by the checker required from any department within RMD or any auxiliary department to be involved as and when required.

8.5.2 Respective key person in charge to prepare a list of obligors tagged with the checker's name tagged.

8.5.3 RMD to ensure that the CRMS ID is created and activated for the users involved.

8.5.4 A traffic light assessment will be conducted on each obligor to indicate the materiality of discrepancy in input of data.

8.5.5 Details for each level of materiality are as follows:-



8.5.6 The checker would perform a second round of validation once the maker has completed the initial round.

8.5.7 The result of data sanity check will be documented for CRO's sign-off and presented to MARCC and BRC.

8.5.8 Efforts to rectify any of the findings shall be escalated to the respective departments with focus on the issues identified as Red.

8.6 Frequency of testing

The data sanity checking process will be implemented on an annual basis.

CHAPTER 9: DATA COLLECTION

9.1 Overview

The main objective is to serve as a reference for the personnel to gather data from the sources and to ensure all information is up to date. The initial point will be the list of existing active obligors of the Bank as of 31st December 2022.

9.2 Purpose

The purpose of the data collection process is to enable the Bank to build up sufficient internal data on credit default and other relevant parameters to allow the bank to conduct an internal analysis and validation of the CRMS system.

9.3 Data Collection

At every quarter, the data collection will be as follows:

i. changes in the portfolio, i.e; new obligors, obligors that have exited relationship with EXIM Bank such as full settlement, prepayment and etc.

ii. changes in the stages of obligors, i.e; obligors that turns into watchlist, obligors declassified as watchlist, obligors that turns into impaired (ECL Stage 3) or declassified from ECL Stage 3 to Stage 1 or Stage 2.

iii. changes in the rates of the obligors, i.e; Total Applied Risk rates, granular rates of Financial (Historical, Projected, Sensitivity), Management, Industry, Environment as well as rates assigned at the metrics level.

iv. changes in the collateral, i.e; the LGD associated with fully secured, partially secured and unsecured.

v. an analysis of the obligor rates transition and the Observed Default Rate (ODR) will be performed amongst others at the end of the fourth (4th) quarter.

9.4 Frequency of Data Collection

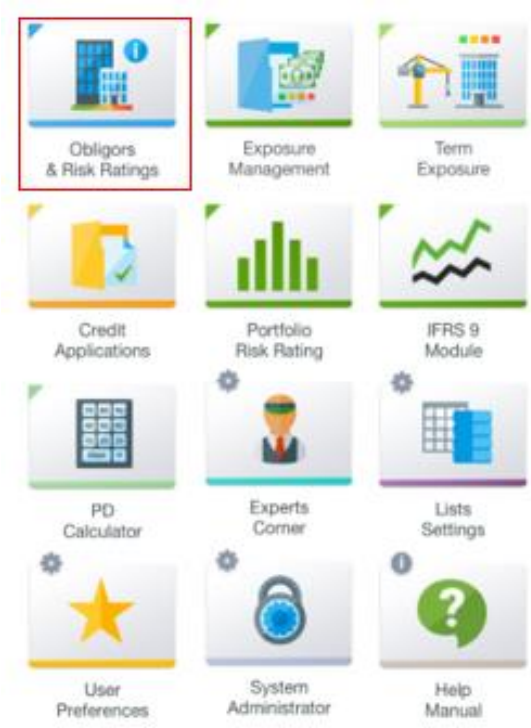
The data collection process will be carried out on a quarterly basis.

9.4 CRMS Process Flow on Data Collection for CRMS Risk Rating

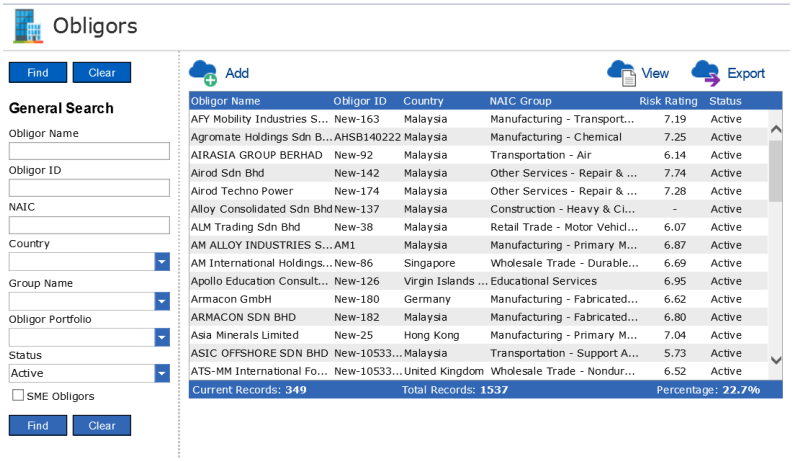
i. Log in to the CRMS account:



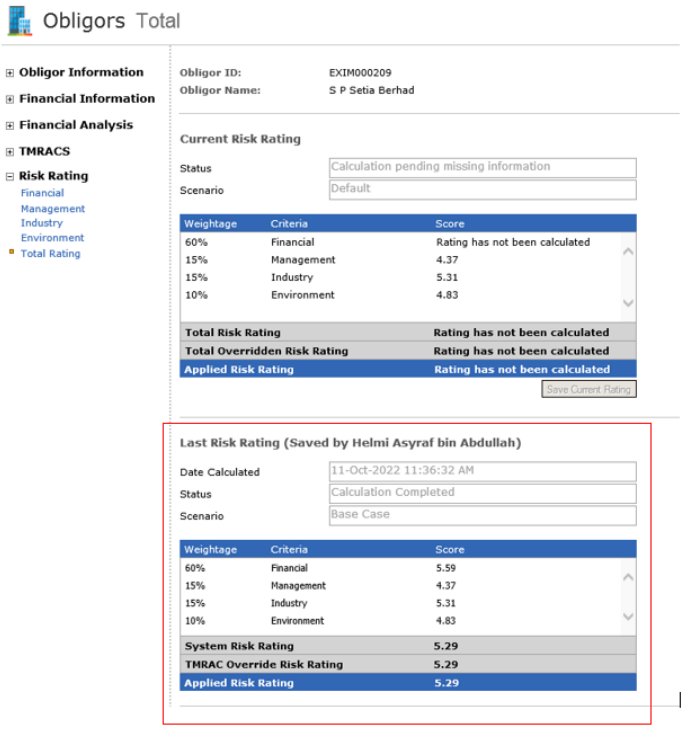
ii. Click on Obligors & Risk Ratings



iii. Key-in and Select Obligors Name



iv. Select Total Rating under the dropdown of Risk Rating



v. Collect the Financial, Management, Industry, Environment and Applied Risk Rating data of the Obligor.

APPENDIX A: Data Dictionary

Data field in the System

No.	Terms in CRMS	Definition
1.	Sales Growth	(Sales year_2 / Sales year_1) -1
2.	COS/Sales	Cost of Sales / Sales
3.	SG&A/Sales	(Selling, General & Administrative Expenses) / Sales
4.	SG&A Growth	(SG&A year_2 / SG&A year_1) -1
5.	EBITDA	Represents Earnings before Interest, Taxes, Depreciation and Amortization and is the result of Section 1 of the Cash Flow Statement.
6.	EBITDA/Sales	(Earnings Before Interest Taxes Depreciation & Amortization) / Sales
7.	EBITDA Tolerance	EBITDA Margin / 8.2% x 30
8.	Net Margin	Net Income / Sales
9.	CAGR	(Ending Value / Beginning Value)^ (1/number of years)-1
10.	Volatility	Stdev (Value Year 1, Value Year 2, …)
11.	Receivable Aging	Trade Receivables/Sales x 365
12.	Receivables + Retention Receivables Aging	(Trade Receivables + Retention Receivables) / Sales x 365
13.	Inventory Aging (COGS)	(Net Inventory + Advances for Inventory) / Cost of Goods Sold x 365
14.	Inventory Aging (Purchases)	(Net Inventory + Advances for Inventory) / Purchases x 365
15.	Inventory + WIP Aging (COGS)	(Net Inventory + Advances for Inventory + WIP) / Cost of Goods Sold x 365
16.	Payable Aging (COGS)	(Trade Payables + Advances from Customers) / Cost of Sales x 365
17.	Payable Aging (Purchases)	(Trade Payables + Advances from Customers) / Purchases x 365
18.	Cash Cycle (COGS)	Receivable Aging + Inventory Aging (COGS) - Payable Aging (COGS)
19.	Cash Cycle (Purchases)	Receivable Aging + Inventory Aging (Purchases) - Payable Aging (Purchases)
20.	Cash Cycle (COGS+WIP)	Receivable Aging + (Inventory Aging + WIP (COGS)) - Payable Aging (COGS)
21.	Cash Cycle + Retention (COGS)	(Receivable + Retention Receivable Aging) + Inventory Aging (COGS) - Payable Aging (COGS)
22.	Cash Cycle + Retention (Purchases)	(Receivable + Retention Receivable Aging) + Inventory Aging (Purchases) - Payable Aging (Purchases)
23.	Cash Cycle + Retention (COGS+WIP)	(Receivable + Retention Receivable Aging) + (Inventory + WIP Aging (COGS)) - Payable Aging (COGS)
24.	Cash Cycle %	Cash Cycle / 30 x 8.2%
25.	Change in Cash Cycle	Cash Cycle(year_2) - Cash Cycle(year_1)
26.	Tolerance less Change in Cash Cycle	EBITDA Tolerance (days) - Change in Cash Cycle

No.	Terms in CRMS	Definition
27.	EBITDA Margin - Cash Cycle %	EBITDA % Sales - Cash Cycle %
28.	Category of Obligor	Cat 1, Cat 2 and Cat3
29.	Cat 1	EBITDA Margin > Cash Cycle %, therefore strong NOCF with Sales Growth. Represents the Obligor that has EBITDA Margin > Cash Cycle % (i.e. the calculation for EBITDA Margin - Cash Cycle % is positive) and has positive NOCF
30.	Cat 2	EBITDA Margin < Cash Cycle % but EBITDA Tolerance > Change in Cash Cycle and Sales Growth reasonable. Therefore NOCF is positive but limited
31.	Cat 3	EBITDA Margin < Cash Cycle % and EBITDA Tolerance < Change in Cash Cycle, and/or NOCF is negative
32.	Funding (COGS)	Cash Cycle (COGS) / 365 x Cost of Goods Sold
33.	Funding (Purchases)	Cash Cycle (Purchases) / 365 x Cost of Goods Sold
34.	Funding + Retention (COGS + WIP)	Cash Cycle + Retention (COGS)) / 365 x Cost of Goods Sold
35.	SG&A Funding per Month	(Total SG&A + Change Prepaids + Change Accruals) / Number of Months
36.	Working Capital	Current Assets - Current Liabilities
37.	Current Ratio	Current Assets / Current Liabilities
38.	Quick Ratio	(Current Assets - Net Inventory) / Current Liabilities
39.	NOCF	Gross Operating Cash Flows + Change in Working Capital Accounts
40.	NOCF Growth	(NOCF Y2 - NOCF Y1) / (NOCF Y1)
41.	NOCF / Sales	(Net Operating Cash Flows) / Sales
42.	Capex / Sales	Capital Expenditure / Total Sales
43.	Sales / Net Fixed Assets	Total Sales / Net Fixed Assets
44.	Depreciation / Net Fixed Assets	Depreciation Expense / Net Fixed Assets
45.	DSCR (Interest)	NOCF / Interest Expense
46.	DSCR (Current Portion of Long-term Debt)	NOCF / Short-term Portion of Long-term Debt
47.	DSCR (Total)	NOCF / (Bank Debt + Short-term Portion of Long-term Debt + Long-term Debt)
48.	Payback Short-term Debt (Years)	(Bank Debt + Short-term Portion of Long-term Debt) / NOCF
49.	Payback Long-term Debt (Years)	Long-term Debt / NOCF
60.	Payback Total Debt (Years)	(Bank Debt + Short-term Portion of Long-term Debt + Long-term Debt) / NOCF
61.	Leverage Total Liabilities	Total Liabilities / Total Equity
62.	Leverage Less Intangibles	Total Liabilities / (Total Equity - Intangible Assets)
63.	Leverage Bank Debt	(Bank Debt + Short-term Portion of Long-term Debt + Long-term Debt) / Total Equity
64.	Leverage Bank Debt Less Intangibles	(Bank Debt + Short-term Portion of Long-term Debt + Long-term Debt) / (Total Equity - Intangible Assets)
65.	Tangible Net Worth	Total Equity - Unrealized Gains/losses - Revaluation Reserves

No.	Terms in CRMS	Definition
66.	Withdrawals / Total Equity	Change in Equity (if negative) / Total Equity
67.	Withdrawals/Tangible Net Worth	Change in Equity (if negative) / Tangible Net Worth
68.	Return on Assets	Net Profits / Total Assets
69.	Return on Equity	Net Profits / Total Equity
70.	Operating Expenses	Net Revenue (net of Cost of Funds) / Revenue to Expense ratio
71.	Exposure at Risk (EAR)	• For Authorized, EAR = Credit Conversion Factor (CCF) x Limits. • For Outstanding's, EAR = CCF x Utilized Amount + CCF for Unutilized x Unutilized Amounts
72.	Obligor Risk Rating (ORR)	Represents the obligor's likelihood to be defaulted within a one-year time horizon and its scales starts from 1 to 10 where '1' is the best rating and '10' is the worst
73.	Facility Risk Rating (FRR)	Risk rating assigned to each individual facility based on the type of facility, the collateral, and the country where the facilities are extended
74.	Portfolio Risk Rating (PRR)	The weighted average of the FRR of the outstanding facilities. The PRR is used to review and manage the overall quality and trajectory of the Bank's credit portfolio. PRR also is an indicator in evaluating the riskiness level of the portfolio at any point in time either from the point view of the obligors, single counterparties, industry or even geography
75.	Target Market and Risk Acceptance Criteria (TMRACS)	Set of assessments that are both numerical and non-numerical in nature that are unique to individual obligors. These are specific to the obligor's sector (e.g. contracting), business (SMEs), facility type, pricing and share of wallet (RACs), and adherence to covenants

APPENDIX B (i): List of Quantitative and Qualitative Items, Analysis & Weightage

Risk Rating	Risk Weightage	Justification for Factor Selection	Justification for Risk Weightage
Financial	**60%**		
Historical	50%	Historical financial data is the most objective and audited information that helps the management committee to make an informed credit decision. Example is to evaluate the cash-generating and paying ability of the borrowers. This is carried out via a set of financial ratios / analysis listed in the table below.	Historical financial performance is the most important consideration given that past performance of the obligor points towards the current financial strength of the obligor in ensuring a sustainable business. E.g., a highly profitable businesses allows the build-up of equity reserves, hence, a stronger balance sheet. A profitable business with healthy cash flow generation capability is often a strong indication of the success and sustainability of the business models. The success of an obligor's business model revolves around the ability to generate Net Operating Cash Flows (NOCF), given that it is used to settle interest charges and short-term obligations towards banks. Hence, from a credit perspective, historical financial performance deserves the highest weightage as compared to other more subjective factors. We have, therefore, set this at 50% of the total financial risk rating given its importance as mentioned above.
Future	33%	Projected financials standing of the borrowers' ability to generate those positive cash flows for their financial obligations.	From credit perspective, historical financial performance must be supported with a view of the future projections of the obligor's business, i.e. whether the projected cash flows can sustain the obligor's business. Projected NOCF and cash cycles are key in determining repayment capability of the obligor. Hence, a third (33%) of the total financial risk rating is assigned to projections to allow a significant consideration of the obligor's outlook. Given that future projections represent the future prospect of the obligor, projections are invariably uncertain in absolute terms, in providing a picture of the obligor's future cash generating ability. Hence, projections must be balanced by Sensitivity Analysis, taking a 17% weightage. In total, Projection (33%) and Sensitivity (17%) accounted for 50% of the financial risk rating which provides an equal balance against the Historical weightage of 50%. We believe the proportion is appropriate to provide a fair view of the financial risk rating of the obligor. The purpose of the sensitivity analysis is to measure the vulnerability of the projections to changes in key factors such as the obligor's future circumstances (major business drivers), and impact on its future cash generating ability. The higher the tolerance to change, the lower is the rating and vice versa. The major business drivers are as follows, with equal distribution of weights into Sales, Cost of Sales I Sales, SG&A/ Sales, Receivable Aging, Inventory Aging, and Payable Aging.
Sensitivity	17%	Represents the measurement of vulnerability to changes in the Obligor's future circumstances and impact on its future cash generating ability.	

Risk Rating	Risk Weightage	Justification for Factor Selection	Justification for Risk Weightage
Management	**15%**	To mitigate any misjudgment of the bank's Relationship Manager in charge of the respective account which can affect the total risk rating significantly.	Management Risk Rating is a qualitative assessment of the obligor based on a series of questions and answers covering a number of key factors: • Character of management • Quality and skills • Organisation structure • Depth and succession • Risk management The number and nature of the questions are important in gauging qualitative issues and reducing, to a high degree, the judgmental element of the assessment. While a 15% weightage for management is not as high as financial risk rating, it still allows a significant consideration of the above management related factors. From credit perspective, a good management is very crucial for the success and sustainability of the business, hence, serve as a complementary support to the financial risk rating.
Industry	**15%**	Collective basis risk rating that based on the independent and detailed research contributed partly by Global Advisory and Research Department and any other approved external rating agencies.	The evaluation is made using a series of questions and answers on the obligor's Industry covering key factors: • Growth Trends and Development Phase • Competitive Environment and Entry Barriers • Capital and Technology Requirements • Regulations The number and nature of the questions are important in gauging qualitative issues and reducing to a high degree the judgmental element of the assessment. A 15% weightage is assigned to allow a significant consideration of the above factors given that from credit perspective, operating in an industry that is growing, with moderate competition and supportive regulations is critical to the top line (revenue/ sales), hence, the obligor's repayment capacity.
Environment	**10%**	Examines external components at a macro-economic level that may pose threats to the business.	The Environment Risk Rating is the third qualitative assessment of the obligor. The evaluation is made using a series of questions and answers on the obligor's environment. As with the other qualitative questions, the number and nature of the questions are important in gauging qualitative issues and reducing to a high degree the judgmental element of the assessment. A 10% weightage is appropriate to balance up the other qualitative dimension of management (15%) and industry (15%). A review of the weightage for Environment can be undertaken in the future once we have sufficient internal default data for model analysis and evaluation.
Total	**100%**		

No	Element	Component	Analysis	Risk Weightage	Justification for Risk Weightage
1	**Financial**				
	Historical	NOCF Volatility	Liquidity	15.00%	NOCF is important from the perspective of repayment capacity of the obligor, hence, a 15% weightage is appropriate.
		DSCR Total Debt	Liquidity	12.99%	DSCR indicates the debt service ability of the borrower and is a common credit indicator in line with industry practices. A 12% weightage for each is appropriate from a credit perspective.
		DSCR Short Term Debt	Liquidity	12.75%	
		NOCF	Liquidity	10.00%	The success of an obligor's business model revolves around the ability to generate Net Operating Cash Flows (NOCF), given that it is used to settle interest charges and short-term obligations towards banks. Hence, a further 10% weightage is deemed acceptable.
		EBITDA/Sales	Profitability	5.00%	EBITDA as a % of Sales indicates the profitability of the business model from the perspective of operating income, hence, sustainability of the business.
		Rec Aging Volatility	Efficiency	5.00%	Cash cycle is key from the perspective of cash generating i.e a healthy cash buffer will allow the obligor to take in higher debt servicing. Cash cycle depends crucially on receivable, inventory and payable days (along with its volatility dimension). Hence, an equal of 5% for each
		Inventory Aging Volatility	Efficiency	5.00%	
		Payable Aging	Efficiency	5.00%	
		Payable Aging Volatility	Efficiency	5.00%	
		EBITDA % - Cash Cycle %	Profitability	5.00%	EBITDA indicates the profitability of the business model from the perspective of operating income, hence, sustainability of the business.
		Vulnerability to Cash Cycle change less change in Cash Cycle	Liquidity	3.00%	Cash Cycle is key from the perspective of cash generating i.e. a healthy cash buffer will allow the obligor to take in higher debt servicing. The vulnerability to changes in Cash Cycle is given a weightage of 3% to incorporate the impact on the financial risk rating od the obligor.

No	Element	Component	Analysis	Risk Weightage	Justification for Risk Weightage
		Sales volatility	Profitability	2.50%	Top line i.e. revenue is the main factor on cash flow. Generally, an obligor with healthy sales will be able to sustain the business model. Topline translates into EBITDA and NOCF eventually. Therefore, 2.5% is appropriate considering the other factors as well.
		EBITDA Volatility	Profitability	2.00%	EBITDA indicates the profitability of the business model from the perspective of operating income, hence, sustainability of the business.
		DSCR (Interest)	Liquidity	2.00%	DSCR indicates the debt service ability of the borrower and is a common credit indicator in line with industry practices. We propose a 2% weightage given the other factors
		Sales growth	Profitability	1.00%	The ability to generate revenue growth is an indicator of the obligor's ability to sustain its business. Hence, a 1% weightage is assigned.
		COS/Sales	Profitability	1.00%	COS/ Sales is an important indicator of the obligor's business margin. This is highly related to profitability, hence, NOCF. Given NOCF and EBITDA have been considered as key factors elsewhere, 1% allows this factor (along with the volatility dimension) to be considered in the overall assessment
		COS/Sales Volatility	Profitability	1.00%	
		SG&A Volatility	Profitability	1.00%	SG&A is the operating expenditure, which is a negative factor in the cash flow (and profitability). 1% is assigned.
		Receivable Aging	Efficiency	1.00%	The Cash Cycle represents the period it takes a company to collect cash from its operations. Hence, a further 1% is allocated to the items as it evaluates the efficiency and liquidity of the business operations.
		Inventory Aging	Efficiency	1.00%	
		Cash Cycle Volatility	Liquidity	1.00%	

No	Element	Component	Analysis	Risk Weightage	Justification for Risk Weightage
		Leverage Total	Solvency	1.00%	Both factors are related to gearing. Highly geared obligors face potential challenges of debt servicing in the event of business/ economy slowdown. a 1% weightage takes these factors into consideration.
		Bank Debt/Equity	Solvency	1.00%	
		SG&A Growth	Profitability	0.50%	SG&A is the operating expenditure, which is a negative factor in the cash flow (and profitability). A significant unjustified growth in SG&A will impact profitability.
		NM Volatility	Profitability	0.10%	Both are important indicators of the obligor's business model's profitability margin (along with the volatility dimension).
		NM % Sales	Profitability	0.05%	
		Current Ratio	Liquidity	0.05%	These are common liquidity indicators. A small weightage is assigned given CRMS placed higher emphasis on cash flow which has been taken into consideration in other metrics.
		Quick Ratio	Liquidity	0.05%	
		Working Capital	Liquidity	0.01%	
	Future	NOCF Volatility	Liquidity	15.00%	As per Historical Risk Weightage
		DSCR Total Debt	Liquidity	12.99%	
		DSCR Short Term Debt	Liquidity	12.75%	
		NOCF	Liquidity	10.00%	
		EBITDA/Sales	Profitability	5.00%	
		Rec Aging Volatility	Efficiency	5.00%	
		Inventory Aging Volatility	Efficiency	5.00%	
		Payable Aging	Efficiency	5.00%	
		Payable Aging Volatility	Efficiency	5.00%	
		EBITDA % - CC %	Profitability	5.00%	
		Vulnerability to CC change less change in CC	Liquidity	3.00%	
		Sales volatility	Profitability	2.50%	
		EBITDA Volatility	Profitability	2.00%	
		DSCR (Interest)	Liquidity	2.00%	
		Sales growth	Profitability	1.00%	
		COS/Sales	Profitability	1.00%	
		COS/Sales Volatility	Profitability	1.00%	
		SG&A Volatility	Profitability	1.00%	
		Receivable Aging	Efficiency	1.00%	
		Inventory Aging	Efficiency	1.00%	

No	Element	Component	Analysis	Risk Weightage	Justification for Risk Weightage
		Cash Cycle Volatility	Liquidity	1.00%	As per Historical Risk Weightage
		Leverage Total	Solvency	1.00%	
		Bank Debt/Equity	Solvency	1.00%	
		SG&A Growth	Profitability	0.50%	
		NM Volatility	Profitability	0.10%	
		NM % Sales	Profitability	0.05%	
		Current Ratio	Liquidity	0.05%	
		Quick Ratio	Liquidity	0.05%	
		Working Capital	Liquidity	0.01%	
	Sensitivity	Change COGS	Profitability	16.67%	The Sensitivity element represents the measurement of vulnerability to changes in the obligor's future circumstances (major business drivers), and impact on its future cash generating ability. The higher the tolerance to change, the better the rating and vice versa. Each major business drivers are equally important and given an equal distribution of weightage.
		Change Sales	Profitability	16.67%	
		Change in Receivables Aging	Efficiency	16.67%	
		Change in Inventory Aging	Efficiency	16.66%	
		Change SGA	Profitability	16.67%	
		Change in Payables Aging	Efficiency	16.66%	

Historical & Future (Financial)			
Liquidity – 10 drivers Total Weightage 56.85%	Profitability – 11 drivers Total Weightage 19.15%	Efficiency – 6 drivers Total Weightage 22.00%	Solvency – 2 drivers Total Weightage 2.00%
Sensitivity (Financial)			
3 Profitability and 3 Efficiency drivers that would ultimately impact the NOCF of the Obligor.			

Justification for Factor Selection

Historical & Future (Financial)

- Liquidity Analysis

  The most significant driver of the CRMS Financial Risk Rating. Reflects the success of the Obligor to generate sufficient cash flow to meet short-term commitments. Liquidity ratios are deemed as an important criterion as not being able to cover interests / instalments indicates a high probability of default. Heavily weighted ratios that relates to liquidity include DSCR, NOCF and Cash Cycle. In addition, traditional liquidity ratios i.e. Current and Quick ratio are also taken into account.

- Profitability Analysis

  Represents the Obligor's operating profitability (EBITDA). All other things equal, the higher the EBITDA margin, the higher is the profitability cushion that allows for greater absorption of changes in working capital accounts (i.e. changes in the Cash Cycle), and therefore the lower is the risk. Analyzing profitability ratios are crucial to assess the sales and expense components of the Obligor before ultimately deriving to the Obligor's profitability.

- Efficiency Analysis

  Measures the current short-term performance of the Obligor by looking at the ability to manage cash flow (receivables / payables) and inventory. A delay or inefficient collection, payment or inventory turnover may result in cash flow problems.

- Solvency Analysis

  Essentially, solvency analysis represents the ability of the Obligor to remain in business in the long future by assessing the total assets against total liabilities and equity. A detailed solvency analysis shall be able to detect early warning flags and uncover the inability of the company to raise further funding, history of financial losses or bad management of funds.

Sensitivity (Financial)

- With respect to the Sensitivity calculation, a given change in any one of the drivers results in a percentage impact on NOCF. However, depending on the level of hedging, risk mitigation and the ability to pass on volatility to customers (captured by responses to "Sensitivity Questions"), a Probability of Impact is calculated ranging from 25%-100%. This Probability is multiplied by the Sensitivity percentage to arrive at a "Probable Impact", or the impact on NOCF after considering this Probability. Using the result (and a rating scale ranging from the lowest risk of 1% probable impact to the highest risk of 100% probable impact), the individual risk rating for each driver is calculated. This rating is then multiplied by the weight applied for each driver (distributed equally across the 6 drivers) to arrive at the Risk Rating score for this section. The summation of these six rating scores results in the Sensitivity Risk Rating.

No	Element	Component	Justification for Factor Selection	Risk Weightage	Justification for Risk Weightage
2	**Management**				
	Character	Integrity	• Management character is of paramount importance to any credit assessment given the sustainability of a business is highly dependent on the management integrity, reliability, and commitment to the success of the company / business. • Such management commitment can be observed / demonstrated via close involvement of the management in day-to-day business operations as well as the management sense of strategy and future direction of the company / business. • Openness / transparency in business dealing is a key indicator of management integrity. Similarly, management unwavering honoring of its obligation is an important indicator of management reliability.	5.56%	The three components of Character carry the same weightage of 5.56% as each item indicates Management's intrinsic value and dedication in steering the business into success and longevity. Strong and robust corporate governance is an important foundation to any business. Absence of such components may lead to loss in investor, supplier, customer or any other relevant stakeholders' confidence which would primarily lead to the fall of the company.
		Reliability		5.56%	
		Commitment		5.56%	
	Quality/Skills	Operational Expertise	• Assessment over Management's expertise and knowledge to lead and manage the business. • This to understand the business rules or "game plan" in the industry and ability to undertake necessary actions to overcome the challenges for survival. • Such management skillsets / expertise should cover the end-to-end process including financial and budgetary control as well as sales and marketing strategy as well as operational execution (professionalism and corporate culture).	4.17%	The four components under Quality / Skills of Management have been assigned a 4.17% weightage. The qualitative assessment on the competency of the Management is given such importance as it looks into the operational ability of Management to lead the business and deliver the required performance for the business to strive. Hence, an evaluation of Management's expertise and experience on the relevant industry or business shall be appraised.
		Financial Control & Expertise		4.17%	
		Marketing Skills		4.17%	
		Professionalism		4.17%	

No	Element	Component	Justification for Factor Selection	Risk Weightage	Justification for Risk Weightage
	Organizational Structure	Hierarchy	• Ability to form management team with relevant experiences and qualifications as to ensure quality decision-making, realistic objective setup and sustainable business strategy. • Human capital and employee's welfare is equally important to the success of the business / project operation. Also, whether appropriate resources and skills are deployed to execute the business plan should be determined. • It is utmost to establish and determine the Ultimate Beneficiary Owner (UBO) as well as the succession plan emplace in the company as part of the AMLA requirement and measures governance structure of the business.	4.17%	A clear definition of authority, responsibility and segregation of duties is essential and shall be assessed under the Organizational Structure element. A well-organized structure lays down the pattern of communication and administration. The strategy and direction of a company can be efficiently disseminated throughout the organization with such effective organizational structure.
		Strategic Management		4.17%	
		Human Resources		4.17%	
		Depth & Succession		4.17%	
	Risk Management	Market Risk	• Identify relevant market risk elements such as FX, interest rate/commission of which may had significant impact toward the business operation and financial capacity and could potentially lead to payment default. • Measure the company relationship and track record of existing facility with respective Bank/FI and compliance toward imposed terms. • Ability of the Management to adapt with the changes in environment and readiness to adjust at consistent or better cost management.	5.56%	Risk Management defines management's ability to identify risks associated with the company and manage or mitigate its exposure to such risks. Market, credit and operational risks are generic risks associated to every business. Hence, a 5.56% weightage is appropriate. Nonetheless, a holistic approach towards the relevant risks to the business shall be further scrutinized.
		Credit Risk		5.56%	
		Operational Risk		5.56%	
	Local/Government Regulations		• Assessment over the company's adherence to the regulatory requirements within the business located in and jurisdictions.	16.67%	A comparably higher weightage is given to Local / Government Regulations amounting 16.67%. as strict adherence to regulatory requirements is essential. Furthermore,

No	Element	Component	Justification for Factor Selection	Risk Weightage	Justification for Risk Weightage
					EXIM Bank priorities and encourages cross-border financing. Hence, failure to comply with other jurisdictions abroad can be highly detrimental to the Bank.
	Relationship with Other Banks		• Determine the Management's ability to diversify its bank facilities and acquire the necessary financial support when needed. • Review conduct of existing account/facilities and security structure offered in obtaining financing.	16.67%	A 16.67% weightage on the review of conduct of account in other banks via relevant credit checking is acceptable as it validates the commitment of the obligor towards it current financial obligations.
3	**Industry**				
	Industry Specific	Growth Trends & Development Stage	• It is important for any credit assessment to consider specific industry factors that are relevant to the borrower's business/project operation. This include the industry development stage i.e., whether the industry is in its infancy, growth, or sunset stage of its lifecycle. Growth stage industry generally are more profitable as compared to sunset stage. Whereas infancy stage generally requires higher Capex. • Other industry specific factors such as the degree of competition as well as any natural or cost type of entry barriers are key in assessing the credit profile of the borrower's business/project operation. Highly competitive industries with low-cost entry barriers are generally less profitable. • In addition to the above, other industry specific factors such as capital and technology requirements as well as any form of regulatory requirements / approval (e.g., banking license) can add to	11.25%	The evaluation on Industry Specifics is made using a series of questions and answers on the Obligor's Industry and is ascertained by EXIM Bank's Global Advisory and Research Department. The assessment broadly covers the important aspects in relation to industry performance at a wholistic level. A 11.25% weightage is distributed equally to the four components covering the common challenges within an industry. Notwithstanding, other components or challenges shall be further assessed.
		Competitive Environment & Entry Barriers		11.25%	
		Capital & Technology Requirements		11.25%	
		Regulations		11.25%	

No	Element	Component	Justification for Factor Selection	Risk Weightage	Justification for Risk Weightage
			further entry barriers, hence, lesser competition that may even result in monopolistic opportunities. Such industries generally are more profitable, hence, translated into better cash flow (better credit profile). • Entry barriers can also be in the form of specific patents, creative talents as well as technological know-hows.		
	Company Specific	Production & Facilities	• Company specific factors are those factors which are considered as idiosyncratic risk to the credit profile of the obligor. • Such company specific factors include production efficiency (economies of scales) and robustness of specific facilities that are supportive of the business growth including capturing market share and leveling competition as well as enhancing market image / reputation (e.g., reputable employers in retaining high skilled talents). Such companies are generally considered of better credit profile given the sustainability of the business in facing market competition. • Other SWOT factors also should be considered in evaluating the credit profile of the borrower including any form of business / product concentration risk (single vs. multiple successful products) as well as technologically superior product that are highly in demand in the market (innovative products). • Overview on products life cycle and its ability to sustain in the market by understanding the critical success factor and competitive edge of the Obligor.	7.50%	45% weightage for Industry is assigned equally to 6 components (7.5%) of Company Specific element that relates to the challenges at company level. The assessment looks into the company's strategy in positioning the business within the industry. Factors that are taken into consideration include but not limited to branding and marketing, market share, product concentration risk, technology, production capacity and etc. A good understanding of the industry allows the business to strategically position itself optimally and enable the business to outperform competitors. Additionally, other strengths and weaknesses of the business are also evaluated to identify and mitigate the relevant risks.
		Market Share & Competitiveness		7.50%	
		Market Image/Reputation		7.50%	
		Concentration Risk		7.50%	
		Technology		7.50%	
		Other SWOT (Strength, Weakness, Opportunities & Threats)		7.50%	

No	Element	Component	Justification for Factor Selection	Risk Weightage	Justification for Risk Weightage
	Sources of Information		• Provide weightage over reliable source of information of which can be verified.	10%	A 10% weightage is assigned to evaluate the reliability of the sources of information given.

No	Element	Justification for Factor Selection	Risk Weightage	Justification for Risk Weightage
4	**Economic**			
	Economic Factor	• Macro-economic environment of the Obligor's country of operation is important from credit assessment perspective given that the borrower are business/project operations is highly dependent on the operating environment such as economic growth (GDP), inflationary pressure (CPI) as well as other factors such as FX/interest rate.	33.33%	Economic factors include minimum wages, interest rates, tax rates governmental activity, and etc. All of these factors occur outside the business but may have a significant impact in the future. Hence, the current and future stability of the economic environment have an acceptable 33.33% weightage to the economic rating of the Obligor.
	Political Factor	• Political Stability of the Obligor's country of operation is relevant to any credit assessment given that any political instability such political coupe, violent protest or even civil war can seriously disrupt the business/project operation of the obligor/borrower.	33.33%	Given EXIM Bank's significant financing exposure towards countries abroad following the Bank's assigned mandate, political factor is placed a significant weightage of 33.33% as it directly impacts the Obligor's ability to complete awarded projects.
	Contagion	• Degree to which the Obligor's environment is impacted by the geopolitical environment of neighboring countries has to be considered in the credit assessment in order to give a holistic picture of the operating environment in which the borrower's business / project operation are located in.	33.33%	Contagion factor is assigned an equal 33.33% weightage as the assessment of geopolitical environment surrounding the country may have a considerable impact on the Obligor's business. All the three Elements under Economic Risk Rating will be derived based on Global Advisory and Research Department's input based on their professional expertise.

APPENDIX B (ii): Scaling – Financial

No	Element	Component	Scoring Methodology	
			Worst	Best
1	**Financial**			
	Historical	Sales CAGR	-10.00%	10.00%
		Sales CAGR Volatility	30.00%	0.00%
		COGS / Sales	100.00%	40.00%
		COGS / Sales Volatility	30.00%	0.00%
		SG&A Fixed CAGR	30.00%	0.00%
		SG&A Fixed CAGR Volatility	30.00%	0.00%
		SG&A Variable % Sales	100.00%	10.00%
		SG&A Variable % Sales Volatility	30.00%	0.00%
		EBITDA/Sales	0.00%	30.00%
		EBITDA/Sales Volatility	30.00%	0.00%
		Net Margins	-10.00%	10.00%
		Net Margins Volatility	30.00%	0.00%
		Receivable Aging	270.00%	15.00%
		Receivable Aging Volatility	30.00%	0.00%
		Inventory Aging	270.00%	15.00%
		Inventory Aging Volatility	30.00%	0.00%
		Payable Aging (COGS)	180.00%	30.00%
		Payable Aging (COGS) Volatility	30.00%	0.00%
		Cash Cycle Volatility	30.00%	0.00%
		EBITDA% - Cash Cycle%	-10.00%	10.00%
		EBITDA% - Cash Cycle% Volatility	30.00%	0.00%
		DSCR total	0.10%	0.70%
		Interest Coverage (NOCF/Interest)	1.60%	0.60%
		DSCR ST Debt	0.40%	1.50%
		Leverage Total	2.00%	0.40%
		Bank Debt/Equity	2.00%	0.40%
		Current Ratio	0.40%	2.00%
		Quick Ratio	0.20%	1.80%
		Working Capital		> 0.00%
	Future	Sales CAGR	-10.00%	10.00%
		Sales CAGR Volatility	30.00%	0.00%
		COGS / Sales	100.00%	40.00%
		COGS / Sales Volatility	30.00%	0.00%
		SG&A Fixed CAGR	30.00%	0.00%
		SG&A Fixed CAGR Volatility	30.00%	0.00%
		SG&A Variable % Sales	100.00%	10.00%
		SG&A Variable % Sales Volatility	30.00%	0.00%
		EBITDA/Sales	0.00%	30.00%
		EBITDA/Sales Volatility	30.00%	0.00%
		Net Margins	-10.00%	10.00%
		Net Margins Volatility	30.00%	0.00%
		Receivable Agin	270.00%	15.00%
		Receivable Aging Volatility	30.00%	0.00%
		Inventory Aging	270.00%	15.00%
		Inventory Aging Volatility	30.00%	0.00%
		Payable Aging (COGS)	180.00%	30.00%
		Payable Aging (COGS) Volatility	30.00%	0.00%
		Cash Cycle Volatility	30.00%	0.00%
		EBITDA% - Cash Cycle%	-10.00%	10.00%
		EBITDA% - Cash Cycle% Volatility	30.00%	0.00%

No	Element	Component	Scoring Methodology	
			Worst	Best
		DSCR total	0.10%	0.70%
		Interest Coverage (NOCF/Interest)	1.60%	0.60%
		DSCR ST Debt	0.40%	1.50%
		Leverage Total	2.00%	0.40%
		Bank Debt/Equity	2.00%	0.40%
		Current Ratio	0.40%	2.00%
		Quick Ratio	0.20%	1.80%
		Working Capital		> 0.00%
	Sensitivity	Change COGS	30.00%	1.00%
		Change Sales	30.00%	1.00%
		Change in Receivables Aging	30.00%	1.00%
		Change in Inventory Aging	30.00%	1.00%
		Change SGA	30.00%	1.00%
		Change in Payables Aging	30.00%	1.00%

APPENDIX B (iii): Mapping from Financial Information into Rating

No	Historical Financial Information	Actual	Weightage	Rating	Weighted Rating
1	Sales CAGR	7%	1.00%	6.50	0.07
2	Sales CAGR Volatility	79%	2.50%	10.00	0.25
3	COGS / Sales	73%	1.00%	5.90	0.06
4	COGS / Sales Volatility	7%	1.00%	3.10	0.03
5	SG&A Fixed CAGR	-7%	0.50%	1.00	0.01
6	EBITDA/Sales	23%	5.00%	3.10	0.16
7	EBITDA/Sales Volatility	8%	2.00%	3.40	0.07
8	Net Margins	11%	0.05%	1.00	0.00
9	Net Margins Volatility	11%	0.10%	4.20	0.00
10	Receivable Aging	93%	1.00%	3.70	0.04
11	Receivable Aging Volatility	38%	5.00%	10.00	0.50
12	Inventory Aging	550%	1.00%	10.00	0.10
13	Inventory Aging Volatility	24%	5.00%	8.20	0.41
14	Payable Aging (COGS)	21360%	5.00%	10.00	0.50
15	Payable Aging (COGS) Volatility	25%	5.00%	8.50	0.43
16	Cash Cycle Volatility	11%	1.00%	4.20	0.04
17	EBITDA% - Cash Cycle%	95%	5.00%	1.00	0.05
18	EBITDA% - Cash Cycle% Volatility	77%	15.00%	10.00	1.50
19	DSCR total	0.07	12.99%	10.00	1.30
20	Interest Coverage (NOCF/Interest)	3.68	2.00%	1.00	0.02
21	DSCR ST Debt	0.28	12.75%	10.00	1.28
22	Leverage Total	1.00	1.00%	4.30	0.04
23	Bank Debt/Equity	0.17	1.00%	1.00	0.01
24	Current Ratio	1.96	0.05%	1.20	0.00
25	Quick Ratio	1.22	0.05%	4.20	0.00
26	Working Capital	528373260%	0.01%	1.00	0.00
	Total Historical Risk Rating				7.08

Appendix C: Risk Rating's Details

(a) Management Risk Rating

No	Question
	Character
	- Integrity
1	Is Management willing to share financial information with the Bank?
2	Has Management ever been inconsistent with the Bank?
3	To what extent does Management show Openness and Transparency in its dealings with the Bank?
	- Reliability
4	Does Management provide accurate information to the Bank?
5	Does Management honor its obligations towards the Bank?
6	Does Management respond to bank requests within a reasonable amount of time?
	- Commitment
7	Management's involvement in day-to-day business matters
8	Management's Personal Attachment to the Company
9	Management Average Compensation
10	Ownership in Company Held by Key Managers
11	Management Sense of Business Strategy (beyond making money for self)
12	Management Content to sacrifice ST personal gains for LT Business gains
13	Management's Genuine Likeness for the Job
	Quality / Skills
	- Operational Expertise
14	To what extent does management have an understanding of its machinery and how it operates?
15	To what extent does management have an understanding of the technologies needed to allow the company to compete?
16	Does management focus on ways to reduce overhead costs and increase efficiency?
17	Does management adequately manage its inventory purchases to minimize either overstocking or inventory deficiencies?
	- Financial Control and Expertise
18	Who is responsible for the financial management of the company?
19	Are budgets and plans set periodically?
20	How often are budgets set?
21	How often are budgets monitored?
22	To what extent are budgets met?
23	Do other functions contribute to the Budget Planning process?
24	Management's primary focus: Operating results or cash flow adequacy?
25	Does the company have a Management/Financial Information System?
26	Can the company's Financial Information System produce accounts quickly and accurately?
27	Can it provide a budget for next month, quarter, fiscal year?
28	Can it produce reports on variations from budget?
29	Are allocations appropriate, followed and monitored?
30	Does it use analytical tools to manage information?
31	Does it produce receivables/payables aging & inform management when accounts are due?
32	Does it use industry gauges to measure status?
33	Do other functions need approval of the assigned Financial Controller in making business decisions?
34	Are the company's financial statements audited?
35	What is the status of the Auditing Firm?
	- Marketing Skills

No	Question
36	How often does management study its existing/potential competition & assess its market share?
37	Does management set departmental sales goals in line with clear market circumstances?
38	To what extent does management have an understanding of its products/services?
39	To what extent is management aware of customer needs?
40	To what extent is management focused on differentiation of its products/services?
41	How effective were the company's marketing and selling strategies in the past?
	- Professionalism
42	Does management portray professionalism in its interaction with the Bank (respecting appointments, speaking cordially, etc)?
43	Does management instill a sense of professionalism within its corporate culture?
44	To what extent does management focus on maintaining a defined company image in its dealings with customers/suppliers/banks?
45	Are there clear-cut & unified procedures in dealing with customers and customer complaints?
	Organization Structure
	- Hierarchy
46	Is there sufficient delegation of authority in the organization?
47	Is there delegation of authority by function? i.e. Operations, Sales, Financial, etc
48	Does management place high importance on Teamwork within and between the various functions of the organization?
49	Does management portray any bureaucracy in its decision-making process?
	- Strategic Management
50	Is the company's mission clearly defined and communicated by management?
51	Are objectives & strategies clearly defined by management?
52	Does management track its progress vis-a-vis the above objectives & strategies?
53	Are objectives & strategies readjusted with changes in the marketplace?
54	Are objectives balanced between Customers, Employees and Shareholders?
55	How well can management adapt to change?
56	Does management have experience during an economic downturn?
	- Human Resources
57	Are management responsibilities clearly defined with written mandates?
58	Are employees responsibilities clearly defined with written mandates?
59	To what extent are employee goals set and performance appraised accordingly?
60	Are quality initiatives set in the employee mandates?
61	Does management set standards & procedures to measure quality?
62	What is the extent of employee turnover in the company?
63	What is the average age of employees in the organization?
64	What are the average years of experience of employees in the organization?
	- Depth & Succession
65	What is the form of ownership in the company?
66	Is actual company management a one-man show?
67	Is the company part of a group of companies? If yes, is there a holding company?
68	Does the company have a large number of transactions with its affiliates?
69	Is there a clearly defined successor for Top Management?
70	Does the company have sufficient back-ups in case of managerial/employee absences?
	Risk Management
	- Market Risk
71	To what extent does management identify the company's foreign exchange and interest rate/commission risks?
72	To what extent does management follow up the markets to anticipate changes in foreign exchange and interest rates/commissions?
73	To what extent does management utilize tools (such as derivatives) to hedge its foreign exchange and interest rate/commission risk?

No	Question
74	To what extent has management successfully mitigated its foreign exchange and interest rate/commission risks in the past?
75	Does management take into account interest rate/commission risk when studying its financing alternatives?
	- Credit Risk
76	In achieving sales, does the company's management take into consideration the amount of exposure to each client?
77	To what extent has management revealed its capability of managing its cash investments?
78	Does management show ability to manage its bank facilities by diversifying its exposure to several banks?
79	To what extent does management properly manage its relationships with suppliers to encourage continuous support & flexibility terms?
	- Operational Risk
80	Does management update its technologies vis-a-vis changes in the industry's technological environment?
81	Is management aware of operational risk?
82	Does management have controls in place to manage its operational risks?
83	To what extent does management manage its operational workflow to minimize the costs of poor scheduling? (operations management)
84	Does management run any quality control or ISO processes?
	Regulations
85	How capable is management in anticipating changes in local/government regulations before they occur?
86	In the past, to what extent has management been able to smoothly adapt to changes in regulations?
	Relationship with Other Banks
87	Are other banks the company deals with reliable enough to support the client in case of need?
88	What share of wallet do other banks enjoy over total facilities?
89	What is the average annual percent utilization over granted facilities?
90	Are there securities offered to banks by the company (including your Bank)?

No	Multiplier Questions
91	Does the Bank have in-depth knowledge of all levels of the company's administration?
92	Did the Bank acquire market's opinion on the company?
93	Does the Bank have contact with senior managers at least once a year?
94	Does the Bank have frequent contact with the company's finance division?
95	Does the Bank have good contacts with other senior management at the company?
96	Is the Bank aware of decisions executed by company's management prior to implementation?
97	Does the Bank have all the needed information from the company?
98	Is there an open relationship between the Bank & the company?

(b) Industry Risk Rating

No	Questions
	Sources of Information
1	To what extent did the bank use Personnel, Management, Top and Bottom Employees to gather information on the company?
2	To what extent did the Bank use Regular Visits To The Site to gather information on the company?
3	To what extent did the Bank use Other Bank Departments (Operations, Trade, Treasury) to gather information on the company?
4	To what extent did the bank use Market References (Clients, Suppliers, Competitors) to gather information on the company?
5	To what extent did the bank use Internet & Other Research to gather information on the company?
6	To what extent did the Bank use Other Banks to gather information on the company?
7	To what extent did the bank use Media & Other Professional Associations to gather information on the company?
	Industry specific
	Growth trends and development stage: Stage of development of the obligor's industry and current / expected growth trends
8	What is the growth trend of the industry?
9	In which stage of development is the industry?
	Competitive environment and entry barriers: Competitiveness of the industry and how easy / difficult it is for new companies to enter and compete within this industry
10	Describe the industry's competitive environment
11	To what extent can new companies enter the industry & succeed?
	Capital and technology requirements: Degree to which investments in capital and technology are required
12	To what extent do firms in this industry have to invest in Property, plant & Equipment?
13	To what extent do firms in this industry have to invest in new technology?
14	Are there patents in this industry to protect company's technologies?
15	To what extent are talent and creativity levels required for this industry?
16	What is the average amount of time in which technology becomes obsolete in this industry?
	Regulations: Regulations imposed on the industry by local governments and the extent to which such regulations change
17	Is the industry exposed to regulations that may negatively impact the operation of companies within the industry?
18	Have the local authorities imposed any regulations or executed any deregulations on the company during the past 5 years?
19	Are the firms in this industry protected by agency rights?
20	Are local products protected from foreign competition through custom duties?
21	Does the company's local country enjoy any Free Trade Agreements?
22	Does the company benefit from any of the free trade agreements with other countries?
23	Does the industry benefit from any government subsidies?
	Company specific
	Production and facilities: Condition of the company's production facilities
24	Introduction Question: Does the company have either a plant or warehouse?
25	Does the company benefit from economies of scale?
26	What is the ratio of management to labor in the company?
	Market share and competitiveness: Company's share of the market and how competitive the operation is relative to other companies within the same industry
27	What market share does the company hold?
28	How competitive are the company's products & services relative to its competition?
29	How easily can the company's products & services be substituted by existing or new competitors?

30	Are products & services delivered in a timely manner?
31	Does the company respond quickly & efficiently to client requests?
No	**Questions**
32	To what extent are customers loyal to the company?
33	What is the quality of the company's after-sale service?
	Market image/reputation: Company's image with its customers, suppliers and employees
34	How do customers perceive the company's products and services?
35	How do employees assess the company in terms of Career Growth Potential?
36	How do employees assess the company in terms of Remuneration & Compensation?
37	How do employees assess the company in terms of Working Conditions & Environment?
38	How do employees assess the company in terms of Discrimination on Race, Gender, Religion, etc?
39	How do employees assess the company in terms of Employee Development: Trainings, Seminars, etc?
40	What reputation does the company hold with its suppliers?
	Concentration risks: Degree of concentration in the customer base, suppliers, and employees
41	How many major suppliers (more than 10% of total purchases) supply the company with inventory/raw material?
42	How many suppliers compose up to 50% of the company's inventory/raw material purchases?
43	How many suppliers compose up to 50% of the company's payables?
44	How would the current terms & conditions offered by the majority of suppliers be characterized?
45	How flexible are the payment terms offered by suppliers?
46	What percentage of purchases is concentrated in one or two foreign currencies?
47	Does the company sell its products/services on a retail basis, wholesale basis or through a 3rd party wholesaler?
48	Does the company rely on its own distribution channels or does it rely on a third party for distribution?
49	Does the company offer its products/services to a broad or narrow segment of the market?
50	How many customers compose up to 50% of the company's receivables?
51	How are the company's credit terms towards its customers characterized?
52	What percentage of sales is to export markets?
53	What percentage of exports is concentrated in one or two countries?
54	How wide is the variety of product/service types offered by the company?
55	Does the company provide a range of products/services within each product/service type?
	Technology: Extent to which technology is invested in and updated as required
56	Introduction Question: Does this business require technology?
57	Does the company invest in R&D to improve its products/services & processes?
58	Does the company benefit from any rights or patents on the technologies it uses?
59	Relative to industry requirements, does the company invest in new technology?
60	Are any of the technologies used by the company considered obsolete?
	Other SWOT: Extent of reliance on resources and raw materials subject to high volatility
61	Are products imported & sold by the company heavily taxed by customs?
62	Is the company dependent on natural resources with high price volatility (such as oil)?
63	Are any of the company's raw materials subjected to significant price changes?
64	Does the company have the ability to transfer such price changes on to its customers?

(c) Environment Risk Rating

No	Questions
	Economic
1	How much is the real GDP growth rate in the country of operation?
2	How much is the level of inflation in the country of operation?
3	What best describes the current state of the economy?
4	What is the unemployment rate in the company's country of operation?
5	Are the country's economic sectors well diversified or are they concentrated in 1-2 major sectors?
6	Is the market subject to local currency devaluations?
	Political
7	Is the country of operation currently facing any political turmoil/unrest?
8	Is the country of operation exposed to any war threats or terror attacks?
9	Are there any street turmoil such as riots, demonstrations, etc?
	Contagion
10	What is the general state of the economy faced by nearby neighboring countries?
11	Are neighboring countries exposed to any war threats or terror attacks?
12	Is there any political turmoil/unrest in the region?
13	Has the country been impacted by the unrest in neighboring countries in the past?

APPENDIX D: Credit Rating's PD Table

RR	PD	RR	PD	RR	PD	RR	PD	RR	PD	RR	PD
1	0.0000%	2	0.0300%	3	0.0600%	4	0.3444%	5	1.3222%	6	3.8772%
1.01	0.0003%	2.01	0.0303%	3.01	0.0603%	4.01	0.3488%	5.01	1.3392%	6.01	3.9146%
1.02	0.0006%	2.02	0.0306%	3.02	0.0606%	4.02	0.3532%	5.02	1.3564%	6.02	3.9526%
1.03	0.0009%	2.03	0.0309%	3.03	0.0609%	4.03	0.3578%	5.03	1.3737%	6.03	3.9909%
1.04	0.0012%	2.04	0.0312%	3.04	0.0612%	4.04	0.3624%	5.04	1.3912%	6.04	4.0297%
1.05	0.0015%	2.05	0.0315%	3.05	0.0615%	4.05	0.3670%	5.05	1.4089%	6.05	4.0689%
1.06	0.0018%	2.06	0.0318%	3.06	0.0686%	4.06	0.3718%	5.06	1.4267%	6.06	4.1086%
1.07	0.0021%	2.07	0.0321%	3.07	0.0709%	4.07	0.3766%	5.07	1.4447%	6.07	4.1488%
1.08	0.0024%	2.08	0.0324%	3.08	0.0732%	4.08	0.3814%	5.08	1.4629%	6.08	4.1895%
1.09	0.0027%	2.09	0.0327%	3.09	0.0755%	4.09	0.3864%	5.09	1.4812%	6.09	4.2307%
1.1	0.0030%	2.1	0.0330%	3.1	0.0778%	4.1	0.3914%	5.1	1.4997%	6.1	4.2724%
1.11	0.0033%	2.11	0.0333%	3.11	0.0801%	4.11	0.3965%	5.11	1.5183%	6.11	4.3147%
1.12	0.0036%	2.12	0.0336%	3.12	0.0825%	4.12	0.4017%	5.12	1.5371%	6.12	4.3575%
1.13	0.0039%	2.13	0.0339%	3.13	0.0848%	4.13	0.4070%	5.13	1.5561%	6.13	4.4009%
1.14	0.0042%	2.14	0.0342%	3.14	0.0872%	4.14	0.4123%	5.14	1.5752%	6.14	4.4449%
1.15	0.0045%	2.15	0.0345%	3.15	0.0896%	4.15	0.4178%	5.15	1.5945%	6.15	4.4895%
1.16	0.0048%	2.16	0.0348%	3.16	0.0919%	4.16	0.4233%	5.16	1.6139%	6.16	4.5347%
1.17	0.0051%	2.17	0.0351%	3.17	0.0943%	4.17	0.4289%	5.17	1.6335%	6.17	4.5805%
1.18	0.0054%	2.18	0.0354%	3.18	0.0967%	4.18	0.4346%	5.18	1.6533%	6.18	4.6270%
1.19	0.0057%	2.19	0.0357%	3.19	0.0992%	4.19	0.4403%	5.19	1.6732%	6.19	4.6742%
1.2	0.0060%	2.2	0.0360%	3.2	0.1016%	4.2	0.4462%	5.2	1.6933%	6.2	4.7221%
1.21	0.0063%	2.21	0.0363%	3.21	0.1040%	4.21	0.4522%	5.21	1.7135%	6.21	4.7707%
1.22	0.0066%	2.22	0.0366%	3.22	0.1065%	4.22	0.4582%	5.22	1.7339%	6.22	4.8200%
1.23	0.0069%	2.23	0.0369%	3.23	0.1089%	4.23	0.4644%	5.23	1.7545%	6.23	4.8701%
1.24	0.0072%	2.24	0.0372%	3.24	0.1114%	4.24	0.4706%	5.24	1.7752%	6.24	4.9210%
1.25	0.0075%	2.25	0.0375%	3.25	0.1138%	4.25	0.4770%	5.25	1.7960%	6.25	4.9727%
1.26	0.0078%	2.26	0.0378%	3.26	0.1163%	4.26	0.4834%	5.26	1.8171%	6.26	5.0252%
1.27	0.0081%	2.27	0.0381%	3.27	0.1188%	4.27	0.4899%	5.27	1.8383%	6.27	5.0785%
1.28	0.0084%	2.28	0.0384%	3.28	0.1213%	4.28	0.4966%	5.28	1.8596%	6.28	5.1327%
1.29	0.0087%	2.29	0.0387%	3.29	0.1238%	4.29	0.5033%	5.29	1.8811%	6.29	5.1879%
1.3	0.0090%	2.3	0.0390%	3.3	0.1263%	4.3	0.5102%	5.3	1.9028%	6.3	5.2439%
1.31	0.0093%	2.31	0.0393%	3.31	0.1288%	4.31	0.5172%	5.31	1.9246%	6.31	5.3009%

RR	PD	RR	PD	RR	PD	RR	PD	RR	PD	RR	PD
1.32	0.0096%	2.32	0.0396%	3.32	0.1314%	4.32	0.5242%	5.32	1.9466%	6.32	5.3588%
1.33	0.0099%	2.33	0.0399%	3.33	0.1339%	4.33	0.5314%	5.33	1.9687%	6.33	5.4178%
1.34	0.0102%	2.34	0.0402%	3.34	0.1365%	4.34	0.5387%	5.34	1.9910%	6.34	5.4778%
1.35	0.0105%	2.35	0.0405%	3.35	0.1390%	4.35	0.5461%	5.35	2.0135%	6.35	5.5389%
1.36	0.0108%	2.36	0.0408%	3.36	0.1416%	4.36	0.5536%	5.36	2.0361%	6.36	5.6010%
1.37	0.0111%	2.37	0.0411%	3.37	0.1441%	4.37	0.5613%	5.37	2.0589%	6.37	5.6643%
1.38	0.0114%	2.38	0.0414%	3.38	0.1467%	4.38	0.5690%	5.38	2.0818%	6.38	5.7287%
1.39	0.0117%	2.39	0.0417%	3.39	0.1493%	4.39	0.5769%	5.39	2.1049%	6.39	5.7943%
1.4	0.0120%	2.4	0.0420%	3.4	0.1519%	4.4	0.5849%	5.4	2.1282%	6.4	5.8611%
1.41	0.0123%	2.41	0.0423%	3.41	0.1545%	4.41	0.5930%	5.41	2.1516%	6.41	5.9292%
1.42	0.0126%	2.42	0.0426%	3.42	0.1571%	4.42	0.6012%	5.42	2.1751%	6.42	5.9985%
1.43	0.0129%	2.43	0.0429%	3.43	0.1597%	4.43	0.6096%	5.43	2.1989%	6.43	6.0692%
1.44	0.0132%	2.44	0.0432%	3.44	0.1624%	4.44	0.6180%	5.44	2.2228%	6.44	6.1412%
1.45	0.0135%	2.45	0.0435%	3.45	0.1650%	4.45	0.6266%	5.45	2.2468%	6.45	6.2146%
1.46	0.0138%	2.46	0.0438%	3.46	0.1676%	4.46	0.6354%	5.46	2.2710%	6.46	6.2894%
1.47	0.0141%	2.47	0.0441%	3.47	0.1703%	4.47	0.6442%	5.47	2.2954%	6.47	6.3657%
1.48	0.0144%	2.48	0.0444%	3.48	0.1730%	4.48	0.6532%	5.48	2.3200%	6.48	6.4435%
1.49	0.0147%	2.49	0.0447%	3.49	0.1757%	4.49	0.6623%	5.49	2.3447%	6.49	6.5228%
1.5	0.0150%	2.5	0.0450%	3.5	0.1784%	4.5	0.6716%	5.5	2.3695%	6.5	6.6038%
1.51	0.0153%	2.51	0.0453%	3.51	0.1811%	4.51	0.6810%	5.51	2.3946%	6.51	6.6863%
1.52	0.0156%	2.52	0.0456%	3.52	0.1838%	4.52	0.6905%	5.52	2.4198%	6.52	6.7706%
1.53	0.0159%	2.53	0.0459%	3.53	0.1865%	4.53	0.7002%	5.53	2.4452%	6.53	6.8565%
1.54	0.0162%	2.54	0.0462%	3.54	0.1893%	4.54	0.7100%	5.54	2.4707%	6.54	6.9442%
1.55	0.0165%	2.55	0.0465%	3.55	0.1920%	4.55	0.7199%	5.55	2.4964%	6.55	7.0337%
1.56	0.0168%	2.56	0.0468%	3.56	0.1948%	4.56	0.7299%	5.56	2.5223%	6.56	7.1251%
1.57	0.0171%	2.57	0.0471%	3.57	0.1976%	4.57	0.7402%	5.57	2.5484%	6.57	7.2184%
1.58	0.0174%	2.58	0.0474%	3.58	0.2004%	4.58	0.7505%	5.58	2.5746%	6.58	7.3136%
1.59	0.0177%	2.59	0.0477%	3.59	0.2032%	4.59	0.7610%	5.59	2.6010%	6.59	7.4108%
1.6	0.0180%	2.6	0.0480%	3.6	0.2061%	4.6	0.7716%	5.6	2.6276%	6.6	7.5101%
1.61	0.0183%	2.61	0.0483%	3.61	0.2089%	4.61	0.7824%	5.61	2.6544%	6.61	7.6114%
1.62	0.0186%	2.62	0.0486%	3.62	0.2118%	4.62	0.7933%	5.62	2.6813%	6.62	7.7150%
1.63	0.0189%	2.63	0.0489%	3.63	0.2147%	4.63	0.8044%	5.63	2.7085%	6.63	7.8207%
1.64	0.0192%	2.64	0.0492%	3.64	0.2176%	4.64	0.8156%	5.64	2.7358%	6.64	7.9287%
1.65	0.0195%	2.65	0.0495%	3.65	0.2206%	4.65	0.8270%	5.65	2.7633%	6.65	8.0390%
1.66	0.0198%	2.66	0.0498%	3.66	0.2235%	4.66	0.8385%	5.66	2.7910%	6.66	8.1518%
1.67	0.0201%	2.67	0.0501%	3.67	0.2265%	4.67	0.8501%	5.67	2.8189%	6.67	8.2669%

RR	PD	RR	PD	RR	PD	RR	PD	RR	PD	RR	PD
1.68	0.0204%	2.68	0.0504%	3.68	0.2295%	4.68	0.8619%	5.68	2.8470%	6.68	8.3846%
1.69	0.0207%	2.69	0.0507%	3.69	0.2325%	4.69	0.8739%	5.69	2.8753%	6.69	8.5048%
1.7	0.0210%	2.7	0.0510%	3.7	0.2356%	4.7	0.8860%	5.7	2.9038%	6.7	8.6277%
1.71	0.0213%	2.71	0.0513%	3.71	0.2387%	4.71	0.8983%	5.71	2.9326%	6.71	8.7532%
1.72	0.0216%	2.72	0.0516%	3.72	0.2418%	4.72	0.9107%	5.72	2.9615%	6.72	8.8815%
1.73	0.0219%	2.73	0.0519%	3.73	0.2449%	4.73	0.9232%	5.73	2.9906%	6.73	9.0126%
1.74	0.0222%	2.74	0.0522%	3.74	0.2481%	4.74	0.9360%	5.74	3.0200%	6.74	9.1466%
1.75	0.0225%	2.75	0.0525%	3.75	0.2513%	4.75	0.9488%	5.75	3.0495%	6.75	9.2836%
1.76	0.0228%	2.76	0.0528%	3.76	0.2545%	4.76	0.9619%	5.76	3.0793%	6.76	9.4236%
1.77	0.0231%	2.77	0.0531%	3.77	0.2578%	4.77	0.9751%	5.77	3.1094%	6.77	9.5667%
1.78	0.0234%	2.78	0.0534%	3.78	0.2611%	4.78	0.9884%	5.78	3.1396%	6.78	9.7130%
1.79	0.0237%	2.79	0.0537%	3.79	0.2644%	4.79	1.0019%	5.79	3.1701%	6.79	9.8626%
1.8	0.0240%	2.8	0.0540%	3.8	0.2678%	4.8	1.0155%	5.8	3.2009%	6.8	10.0154%
1.81	0.0243%	2.81	0.0543%	3.81	0.2712%	4.81	1.0294%	5.81	3.2319%	6.81	10.1717%
1.82	0.0246%	2.82	0.0546%	3.82	0.2746%	4.82	1.0433%	5.82	3.2632%	6.82	10.3315%
1.83	0.0249%	2.83	0.0549%	3.83	0.2781%	4.83	1.0575%	5.83	3.2947%	6.83	10.4948%
1.84	0.0252%	2.84	0.0552%	3.84	0.2816%	4.84	1.0717%	5.84	3.3265%	6.84	10.6618%
1.85	0.0255%	2.85	0.0555%	3.85	0.2852%	4.85	1.0862%	5.85	3.3585%	6.85	10.8326%
1.86	0.0258%	2.86	0.0558%	3.86	0.2888%	4.86	1.1008%	5.86	3.3909%	6.86	11.0071%
1.87	0.0261%	2.87	0.0561%	3.87	0.2924%	4.87	1.1156%	5.87	3.4235%	6.87	11.1856%
1.88	0.0264%	2.88	0.0564%	3.88	0.2961%	4.88	1.1305%	5.88	3.4564%	6.88	11.3681%
1.89	0.0267%	2.89	0.0567%	3.89	0.2998%	4.89	1.1456%	5.89	3.4896%	6.89	11.5547%
1.9	0.0270%	2.9	0.0570%	3.9	0.3036%	4.9	1.1608%	5.9	3.5231%	6.9	11.7454%
1.91	0.0273%	2.91	0.0573%	3.91	0.3075%	4.91	1.1762%	5.91	3.5570%	6.91	11.9404%
1.92	0.0276%	2.92	0.0576%	3.92	0.3113%	4.92	1.1918%	5.92	3.5912%	6.92	12.1398%
1.93	0.0279%	2.93	0.0579%	3.93	0.3153%	4.93	1.2075%	5.93	3.6256%	6.93	12.3437%
1.94	0.0282%	2.94	0.0582%	3.94	0.3193%	4.94	1.2234%	5.94	3.6605%	6.94	12.5521%
1.95	0.0285%	2.95	0.0585%	3.95	0.3233%	4.95	1.2395%	5.95	3.6957%	6.95	12.7653%
1.96	0.0288%	2.96	0.0588%	3.96	0.3274%	4.96	1.2557%	5.96	3.7312%	6.96	12.9832%
1.97	0.0291%	2.97	0.0591%	3.97	0.3316%	4.97	1.2721%	5.97	3.7671%	6.97	13.2059%
1.98	0.0294%	2.98	0.0594%	3.98	0.3358%	4.98	1.2886%	5.98	3.8034%	6.98	13.4337%
1.99	0.0297%	2.99	0.0597%	3.99	0.3400%	4.99	1.3053%	5.99	3.8401%	6.99	13.6666%

APPENDIX E: The one-year Transition Matrix

Table 21

Global Corporate Average Transition Rates (1981-2021)

(%)

From/to	AAA	AA	A	BBB	BB	B	CCC/C	D	NR	
1-year										
AAA	87.09	9.05	0.53	0.05	0.11	0.03	0.05	0.00	3.10	100.01
	(7.20)	(7.22)	(0.82)	(0.24)	(0.27)	(0.17)	(0.34)	(0.00)	(2.43)	
AA	0.48	87.32	7.72	0.46	0.05	0.06	0.02	0.02	3.88	100.01
	(0.53)	(5.16)	(4.11)	(0.67)	(0.19)	(0.20)	(0.06)	(0.07)	(1.79)	
A	0.02	1.56	88.73	4.97	0.25	0.11	0.01	0.05	4.29	99.99
	(0.09)	(1.06)	(3.76)	(2.12)	(0.38)	(0.24)	(0.06)	(0.10)	(1.67)	
BBB	0.00	0.08	3.19	86.72	3.48	0.42	0.09	0.15	5.86	99.99
	(0.03)	(0.15)	(1.61)	(3.96)	(1.58)	(0.64)	(0.20)	(0.24)	(1.53)	
BB	0.01	0.02	0.10	4.52	78.12	6.66	0.53	0.60	9.43	99.99
	(0.05)	(0.08)	(0.23)	(1.97)	(4.54)	(3.10)	(0.69)	(0.80)	(2.16)	
B	0.00	0.02	0.06	0.15	4.54	74.73	4.81	3.18	12.51	100
	(0.00)	(0.08)	(0.19)	(0.20)	(2.15)	(3.91)	(2.76)	(3.01)	(2.24)	
CCC/C	0.00	0.00	0.09	0.16	0.49	13.42	43.91	26.55	15.39	100.01
	(0.00)	(0.00)	(0.38)	(0.59)	(0.85)	(7.61)	(8.40)	(11.94)	(4.70)	

Remarks: To refer S&P Global Rating Transition Rate

Initial Transition Matrix

Initial Rating	1.00	2.00	3.00	4.00	5.00	6.00	7.00	8.00
1	87.09	9.05	0.53	0.05	0.11	0.03	0.05	3.10
2	0.48	87.32	7.72	0.46	0.05	0.06	0.02	3.90
3	0.02	1.56	88.73	4.97	0.25	0.11	0.01	4.34
4	0.00	0.08	3.19	86.72	3.48	0.42	0.09	6.01
5	0.01	0.02	0.10	4.52	78.12	6.66	0.53	10.03
6	0.00	0.02	0.06	0.15	4.54	74.73	4.81	15.69
7	0.00	0.00	0.09	0.16	0.49	13.42	43.91	41.94
8	0.00	0.00	0.00	0.00	0.00	0.00	0.00	100.00
9	0.00	0.00	0.00	0.00	0.00	0.00	0.00	100.00
10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	100.00

Remarks: For initial rating 8.00 = To sum up the rate S&P from columns D and NR. It is to ensure 100% summation over the remaining ratings.


EXIM
BANK
MALAYSIA